

National Guideline for Risk Assessment and Primary Prevention of Cardiovascular Diseases For Secondary and Tertiary Healthcare level

**Directorate of Non-Communicable Diseases
Ministry of Health
2021**



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National Guideline for Risk Assessment and Primary Prevention of Cardiovascular Diseases
For Secondary and Tertiary healthcare level

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This document was reviewed by the Directorate of NCD to be in line with the National policies, strategies and regulations.

Feedback of relevant professional colleges and institution was also incorporated into this guideline

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Abbreviations

ABI	Ankle-brachial index
ACE-I	Angiotensin converting enzyme inhibitors
ApoA	Apolipoprotein A
ApoB	Apolipoprotein B
ARB	Angiotensin receptor blockers
ASCVD	Atherosclerotic cardiovascular disease
AU	Agatston score
BMI	Body mass index
CAD	Coronary artery disease
CCS	Coronary calcium score
CKD	Chronic kidney disease
COR	Class of recommendation
CTCA	Computerized Tomography Coronary Angiography
CVD	Cardiovascular disease
DBP	Diastolic blood pressure
ECG	Electrocardiogram
ED	Erectile dysfunction
GLP-1R	Glucagon-like peptide-1 receptor
HDL	High density lipoprotein
hsCRP	High-sensitivity C-reactive protein
IMT	Intima-media thickness
ISH	International Society of Hypertension
LDL	Low density lipoprotein
Lp(a)	Lipoprotein(a)
LOE	Level of evidence
LVH	Left ventricular hypertrophy
MPI	Myocardial perfusion imaging
MRI	Magnetic resonance imaging
NNT	Number needed to treat
NRT	Nicotine replacement therapy
OSA	Obstructive sleep apnea
PET	Positron emission tomography
SBP	Systolic blood pressure
SGLT-2	Sodium-glucose cotransporter 2
SPECT	Single-photon emission computed tomography
WHO	World Health Organization
WHR	Waist to hip ratio

Introduction

This guideline is an attempt to translate scientific evidence into clinical practice with a view to improve cardiovascular health in Sri Lankan population.

This guideline attempts to help healthcare practitioners in cardiovascular risk assessment on most patients in most situations. However, the final judgment on assessment procedures and investigations for a particular patient must be made by the healthcare practitioner and it may be appropriate to have deviations from these guidelines on special circumstances.

Atherosclerotic cardiovascular disease (ASCVD) is the leading cause of death worldwide. A number of studies have suggested that South Asian (India, Pakistan, Bangladesh and Sri Lanka) patients have a higher incidence of ASCVD than Caucasian patients.^{1,2} ASCVD is the leading cause of death in Sri Lanka.³ The life expectancy is estimated to increase by almost 7 years if all forms of major ASCVD were eliminated.⁴ Coronary artery disease (CAD) has a long asymptomatic latent period, which provides an excellent opportunity for early detection and prevention. There is established evidence by meta-analysis of randomized clinical trials that absolute risk reduction is larger when individuals with higher baseline risk are detected and treated with antihypertensive medications.^{5,6,7} The number of individuals need to treat (NNT) to prevent one adverse outcome decreases with increasing baseline ASCVD risk.

The aim of this guideline is to provide an evidence-based approach to risk assessment with a view to reduce ASCVD related morbidity and mortality in asymptomatic adults. Every effort was made to keep the guidelines concise and user friendly.

Assessment of the ASCVD risk for Sri Lankan adults should be started from the age of 20 years considering the incidence of ASCVD outcomes in young adults in Sri Lanka and in other South Asian countries. From age 20 to 39 years, it may be reasonable to assess conventional ASCVD risk factors every 4-6 years.^{8,9} However, this risk should be assessed routinely from age 40 to 75 years.⁵

The recommendations published in this guideline are based on established scientific evidence as much as possible. An extensive literature search on published clinical trials, meta-analysis, case studies, consensus opinions and standards of care on cardiovascular risk assessment were reviewed during August to November 2020. The review was limited to the articles published in English language and research done on human subjects.

The key words used for the literature search included ankle-brachial index, apolipoproteins, arterial stiffness, atrial fibrillation, autoimmune diseases, cardiovascular risk, carotid intima-media thickness, carotid ultrasound scans, chemotherapy, chronic kidney disease, computed tomography coronary angiography, coronary calcium score, diabetes, erectile dysfunction, exercise electrocardiogram, genotype testing, global risk scores, hemoglobin A1c, high-sensitivity C-reactive protein, hypercholesterolemia, hyperlipidemia, hypertension, lipoprotein-associated phospholipase A2, lipoproteins, magnetic resonance imaging of plaques, microalbuminuria, myocardial perfusion imaging, natriuretic peptides, obstructive sleep apnea, radiotherapy, resting electrocardiogram, smoking, stress echocardiography, transthoracic echocardiography and weight control.

Scope of the Guidelines

The main focus of this guideline is risk assessment and primary prevention of CVD in asymptomatic men and women to prevent cardiovascular outcomes. These cardiovascular outcomes include CAD (chronic stable angina, unstable angina, coronary revascularization and fatal and non-fatal myocardial infarction), heart failure and atrial fibrillation.

Class of recommendation (COR)

The writing committee performed a formal literature review and weighed the strength of evidence for and against particular procedures, investigations and expected health outcomes. The strengths of evidence for efficacy of interventions were classified as follows.

- **Class I** – Strong evidence to confirm that the benefit of the intervention is much more than the risk.
- **Class IIa** – Moderately strong evidence to suggest that the benefit of the intervention is more than the risk.
- **Class IIb** – Weak evidence to suggest that the benefit of the intervention is more than the risk.
- **Class III (No Benefit)** – Either no evidence to confirm a benefit or the benefit is equal to the risk.
- **Class III (Harm)** – Strong evidence to confirm harm or the risk is more than the benefit.

Level of Evidence (LOE)

The review included ranking of the weight evidence to support recommendations. The evidence was ranked as follows.

- **Level A** - if the data were derived from multiple randomized clinical trials or meta-analyses.
- **Level B** - when the evidence was derived from a single randomized trial or non-randomized studies.
- **Level C** - when the recommendations are based on consensus opinion, retrospective studies, registries, case studies or standard of care.

Key Message

Individuals who have a low-risk for clinical events are unlikely to gain a substantial benefit from pharmacological therapy or revascularization procedures and therefore best be managed with lifestyle modifications only. Individuals who have a high-risk for clinical events are more likely to benefit from pharmacological therapy or revascularization procedures and therefore best managed with such treatment methods. Investigations for cardiovascular risk assessment may be indicated for individuals at borderline intermediate risk and intermediate risk to assess the need for lifestyle modifications and pharmacological therapy.

Chapter 1

Cardiovascular Risk Assessment

1.1 Global Risk Score (GRS) Assessment

Non-modifiable risk factors such as age and gender and modifiable risk factors such as smoking status, diabetes, systolic blood pressure or treatment for hypertension and serum total cholesterol levels can be combined together to develop a predictive model to estimate the risk of morbidity and mortality related to ASCVD.¹⁰ These scores combine multiple risk factors into a single quantitative risk estimate.¹¹ There are many global risk assessment scores available and are used all over the world. The writing committee recommends to use WHO (World Health Organization) risk prediction chart of 2019 for South East Asia for global risk score calculation for Sri Lankan population.

WHO CVD risk prediction chart for South East Asia is specially designed to assess 10-year combined myocardial infarction and stroke (fatal and non-fatal) in South East Asian countries including Sri Lanka.¹² This risk score is a product of a multiple ASCVD risk factors including gender, age, systolic blood pressure, total blood cholesterol, smoking status and presence or absence of diabetes. However, like many other charts, this chart cannot be used to assess GRS in young individuals < 40 years of age.

Other risk calculators, such as ASCVD Risk Estimator¹³ by American College of Cardiology and American Heart Association and Systematic Coronary Risk Estimation (SCORE)¹⁴ used by European Society of Cardiology, are also widely used in Sri Lanka for ASCVD risk assessment. However, ASCVD Risk Estimator and SCORE may predict lower risk estimations for South Asians as they are primarily established for risk assessment in American and European populations respectively. Therefore, correction factors should be applied for more realistic risk estimation when these are used for South Asian populations. For example, SCORE risk estimation should be multiplied by 1.4 for correction to South Asian populations.¹⁵

Individuals can be categorized into low, intermediate and high-risk according to their global risk estimates of fatal or non-fatal outcomes from CVD during next 10 years. Traditionally, GRS of 0-10% are considered as low-risk, 10-20% as intermediate risk and > 20% as high-risk.¹⁶

The GRS assessments should be interpreted together with the knowledge and the experience of the clinician and in view of the factors that may modify the ASCVD risk. Even if the 10-year absolute risk may be low, the relative risk can be high in young persons as events usually occur later in the life.

Table 1. 1 Global Risk Score (GRS) Assessment

COR	LOE	Recommendation
I	C	Calculation of GRS (such as WHO CVD risk prediction chart for Southeast Asia) using multiple, traditional cardiovascular risk factors should be done for all asymptomatic men and women above 40 years of age and have no clinical history of ASCVD.

1.2 Obtaining Family History of Premature ASCVD

Family history of premature ASCVD is an independent risk factor for development of the same disease in asymptomatic men and women.¹⁷ The association of family history may be both the genetic trait and the environment share among family members.¹⁸ Premature ASCVD is defined as a cardiovascular event occurring in a first-degree male relative of age < 55 years and/or in a first-degree female relative of age < 65 years. A family history of premature ASCVD is a simple, inexpensive but valuable information and therefore should be included in cardiovascular risk assessment in all individuals.

Table 1. 2 Obtaining Family History of Premature ASCVD

COR	LOE	Recommendation
I	C	Family history of premature ASCVD should be obtained for risk assessment in all asymptomatic men and women.

1.3. Risk Assessment in Younger Individuals

Individuals < 40 years demonstrated a low ASCVD risk whenever 5- or 10-year their risk is calculated, using a standard risk calculator, regardless of their underlying risk factors. However, some individuals have high relative risk and are likely to experience fatal or non-fatal ASCVD events prematurely.

All the people < 40 years should be evaluated for cardiac risk factors. Men and women with a family history of premature CAD should be investigated for primary hypercholesterolemia using a validated clinical score.^{19,20,21} They should also be evaluated every 4-6 years for other risk factors including smoking status, blood pressure and HbA1c.

Table 1. 3 Risk Assessment in Younger Individuals

COR	LOE	Recommendation
I	B	Screening of all men and women with a family history of premature ASCVD in a first degree relative is recommended for familial hypercholesterolaemia.

1.4. Assessment of Psychosocial Factors

Numerous psychosocial factors are associated with increased risk of ASCVD. These factors include low socio-economic status, chronic stress, mental disorders, personality traits and lack of social support. Poor adherence to treatment for risk factor control and lack of efforts to lifestyle changes may play a role in increased ASCVD risk in this group of individuals.

Low socio-economic status including low educational level, low job status, low income and living in poor residential facilities are associated with increased ASCVD risk.^{22,23}

Chronic stress at work²⁴ and family life²⁵ are also associated with increased ASCVD risk. Also, clinical depression,²⁶ anxiety,²⁷ and schizophrenia²⁸ are associated with increased ASCVD risk. Individuals with hostility, which is characterized by mistrust, rage and anger are also associated with increased ASCVD risk.²⁹

Social isolation and lack of social support are other psychosocial factors associated with increased ASCVD risk.³⁰

Table 1. 4 Assessment of Psychosocial Factors

COR	LOE	Recommendation
IIa	B	Assessment of psychosocial factors in an interview by a clinician or using a standardized questionnaire ³¹ may be helpful for ASCVD risk assessment in all asymptomatic men and women.

1.5. Female Specific Conditions

Pre-eclampsia occurs in about 1-2% while pregnancy-induced hypertension occurs in about 10-15% of all the pregnancies as obstetric complications. Pre-eclampsia is associated with increased future risk of hypertension,³² diabetes³³ and ASCVD³³ while pregnancy-induced hypertension is associated with increased future risk of hypertension³² and ASCVD.³⁴

Gestational diabetes is associated with increased risk of developing diabetes with up to 50% of them develop diabetes within the first 5 years after the pregnancy.³⁵

In a systematic review and a meta-analysis, women who experienced menopause younger than 45 years and women 45 years or older at onset were compared for CVD related outcomes.³⁶ The relative risks of overall CAD, fatal CAD, overall stroke, stroke mortality, CVD mortality and all-cause mortality were increased modestly.³⁶

Polycystic ovarian syndrome is associated with increased future risk of developing diabetes³⁷ and may also be associated with increased ASCVD risk.

Table 1. 5 Female Specific Conditions

COR		LOE	Recommendation
IIa		B	Periodic screening for hypertension and diabetes mellitus can be helpful in women who developed pre-eclampsia, pregnancy-induced hypertension and premature menopause (≤ 40 years).
IIa		B	Periodic screening for diabetes mellitus can be helpful in women who has polycystic ovarian syndrome.

1.6. Chronic Kidney Disease (CKD)

CKD is associated with increased prevalence of hypertension, diabetes and hypercholesterolemia. CKD is also associated with increased risk of ASCVD, independent of conventional cardiac risk factors.³⁸ This increased risk could be due to vascular endothelial inflammation and coronary calcification caused by CKD.³⁸ As a result, eGFR is associated reciprocally with ASCVD. The ASCVD risk can be classified as "very high" for severe CKD (eGFR < 30 mL/min/1.73m²) and as "high risk" for moderate CKD (eGFR < 30-59 mL/min/m²) patients.

Table 1. 6 Chronic Kidney Disease (CKD)

COR	LOE	Recommendation
IIa	B	Measurement of eGFR can be useful for ASCVD risk assessment in asymptomatic men and women with chronic kidney disease.

1.7. Detection of Microalbuminuria

Urine analysis for microalbuminuria is a simple, inexpensive and widely available test, which can be helpful to assess ASCVD risk of both diabetic and nondiabetic patients.³⁹ An early morning, spot urine sample to measure urine albumin-to-creatinine ratio is effective in assessing ASCVD risk.⁴⁰ Microalbuminuria is defined as urine albumin-to-creatinine ratio of 30-300 mg/g.⁴¹

Microalbuminuria is an independent risk factor that is associated with doubling the risk of ASCVD when compared to the individuals without microalbuminuria.⁴² This increased risk was demonstrated across subgroups of individuals including those with and without hypertension, with and without diabetes and with and without low estimated glomerular filtration rate.^{39,41,42}

Individuals at intermediate-risk of ASCVD with microalbuminuria had a substantially high 5-year risk of ASCVD when compared to the individuals without microalbuminuria (20.1% versus 6.3% respectively).⁴³

Table 1. 7 Detection of Microalbuminuria

COR	LOE	Recommendation
IIa	B	Detection of microalbuminuria by urine analysis can be useful for ASCVD risk assessment in asymptomatic men and women with diabetes and hypertension.
IIb	B	Detection of microalbuminuria by urine analysis may be useful for ASCVD risk assessment in asymptomatic men and women at intermediate risk without diabetes and hypertension.

1.8. Measurement of High-Sensitivity C-Reactive Protein (hsCRP)

Inflammation of the vascular endothelium plays a significant role in the pathogenesis of atherosclerosis. As a result, numerous biomarkers have been evaluated as risk markers of ASCVD. hsCRP is the most intensively studied biomarker for inflammation of vascular endothelium.

hsCRP is an independent risk factor for ASCVD.^{44,45,46} hsCRP is also associated with increased risk for stroke, peripheral vascular disease, heart failure, atrial fibrillation, sudden cardiac death and all-cause mortality.^{47,48,49,50,51}

Table 1. 8 Measurement of High-Sensitivity C-Reactive Protein (hsCRP)

COR	LOE	Recommendation
IIa	B	Measurement of hsCRP can be helpful for ASCVD risk assessment in men ≥ 50 years and women ≥ 60 years with no history of CAD, CKD, diabetes mellitus, severe inflammatory conditions and serum LDL cholesterol < 130 mg/dL with no contraindication to statin therapy.
IIb	B	Measurement of hsCRP may be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk, men ≤ 50 years and women ≤ 60 years.
III No Benefit	B	Measurement of hsCRP is not recommended for ASCVD risk assessment in asymptomatic, high-risk, men and women.
III No Benefit	B	Measurement of hsCRP is not recommended for ASCVD risk assessment in asymptomatic, low-risk, men ≤ 50 years and women ≤ 60 years.

1.9. Coronary Calcium Score (CCS)

Although coronary artery atherosclerosis does not always show calcification, it may occur in late-stage subclinical coronary atherosclerosis.⁵² The extent of coronary calcifications correlates with extent of atherosclerotic plaque burden.⁵² The amount of calcium in the coronary artery walls is quantified by measuring the area of the calcium deposits on the scan multiplied by a weighing factor depending on the Hounsfield unit density of the calcium deposits.⁵³ The total amount of coronary calcium is usually expressed as Agatston score (AU).⁵⁴ The radiation dose is low (0.9-1.1

mSv) when the acquisition is prospectively triggered^{55,56} but the radiation dose is higher when the retrospective imaging is used.⁵⁷

Many published studies have reported that elevated CCS provides information on ASCVD risk over and above the information provided by conventional risk factors. Intermediate-risk patients with elevated CCS > 300 AU had 2.8% annual rate of fatal and non-fatal myocardial infarctions (roughly equal to 10-year risk of 28%).⁵³ Therefore, measurement of CCS may be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk and borderline intermediate-risk, men and women.⁵⁸

Measurement of CCS is not recommended for men < 40 years and women < 50 years considering the radiation dose and very low prevalence of detecting coronary calcium in these age groups.^{58,59}

Table 1. 9 Coronary Calcium Score (CCS)

COR	LOE	Recommendation
IIa	B	CCS can be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk, men and women.
IIb	B	CCS can be helpful for ASCVD risk assessment in asymptomatic, low to intermediate-risk (6%-10% 10-year risk), men and women.
III No Benefit	B	CCS is not recommended for ASCVD risk assessment in asymptomatic, low- risk, men and women.

1.10. Patients with Autoimmune Diseases

Active rheumatoid arthritis is associated with increased ASCVD risk in both men and women.⁶⁰ Other autoimmune diseases such as psoriasis,⁶¹ ankylosing spondylitis and systemic lupus erythematosus are also associated with increased ASCVD risk. Systemic inflammation may be involved with accelerated atherosclerosis.⁶² These inflammatory autoimmune diseases increase the relative risk of ASCVD by about 1.4 in men and 1.5 in women.⁶³

Table 1. 10 Patients with Autoimmune Diseases

COR	LOE	Recommendation
IIa	B	The ASCVD risk can be calculated by multiplying the GRS by 1.5 in patients with active rheumatoid arthritis.
IIb	C	The ASCVD risk may be calculated by multiplying the GRS by 1.5 in patients with active non-rheumatoid arthritis conditions.

1.11. Erectile Dysfunction (ED)

Erectile dysfunction is common and may occur up to 40% of men after 40 years of age. ED and ASCVD share common risk factors including diabetes, hypertension, hypercholesterolemia, smoking and depression. Also, these two conditions share the same pathophysiologic mechanisms. Many studies have established that ED is associated with asymptomatic ASCVD in men.^{64,65} A meta-analysis has established that ED is associated with 44% higher risk of ASCVD, 62% higher risk of myocardial infarction, 39% higher risk of stroke and 25% higher risk of all-cause mortality when compared with men without ED.⁶³

Table 1. 11 Erectile Dysfunction (ED)

COR	LOE	Recommendation
Ila	C	Assessment for risk cardiac factors, and symptoms and signs of heart disease can be helpful for ASCVD risk assessment in men with ED.

1.12. Ankle-Brachial Index (ABI)

Ankle-Brachial Index is an easy-to-perform, bed side measurement performed by Doppler measurement of blood pressure in four extremities. It is the value of the highest blood pressure in lower extremity divided by highest blood pressure of upper extremity.

ABI < 0.9 is a reliable marker of peripheral arterial disease.⁶⁶ Many studies have demonstrated that ABI is also inversely related to the ASCVD risk.⁶⁷ Also, studies have demonstrated that measuring a low ABI of < 0.9 has an additive predictive value over GRS assessment.^{68,69}

Table 1. 12 Ankle-Brachial Index (ABI)

COR	LOE	Recommendation
IIb	B	Measurement of ABI may be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk, men and women.

1.13. Carotid Ultrasound

Evidence of atherosclerosis in one arterial territory may be associated with involvement of other arteries.⁷⁰ This can be used as a strategy by ultrasonographic assessment of carotid arteries to predict coronary atherosclerosis. Carotid ultrasonography can be used to measure intima-media thickness (IMT) and to determine presence and characteristics of atherosclerotic plaques.

Carotid IMT includes measurement of early atherosclerosis as well as smooth muscle hypertrophy/hyperplasia. However, lack of having a standardized definition for the measurement, high variability and low reproducibility of the measurement have made carotid IMT failing to demonstrate a value in ASCVD risk prediction.⁷¹ Therefore, use of carotid IMT measurement for ASCVD risk prediction is not recommended.

Carotid ultrasound is used to detect atherosclerotic plaques causing > 50% stenosis. These plaques are associated with both coronary and cerebrovascular events.⁷² Therefore, carotid artery plaque assessment by ultrasonography may be helpful for ASCVD risk prediction.

Table 1. 13 Carotid Ultrasound

COR	LOE	Recommendation
IIb	B	Carotid artery plaque assessment on ultrasonography may be helpful in ASCVD risk assessment in asymptomatic men and women.
III No Benefit	A	Measurement of carotid artery IMT is not recommended for ASCVD risk assessment in asymptomatic men and women.

1.14. Resting Electrocardiogram (ECG)

Population studies have shown that resting ECG abnormalities are predictors of ASCVD risk among asymptomatic individuals.⁷³ Novacode criteria divide resting ECG abnormalities in to major and minor types.⁷⁴ Major abnormalities include atrial fibrillation, atrial flutter, complete atrio-ventricular block, LVH with repolarization abnormalities, isolated ischemic changes and ventricular or supra ventricular tachycardia. Minor abnormalities include first and second degree atrio-ventricular blocks, partial bundle branch blocks and isolated minor Q wave and ST-T abnormalities.

Evidence of LVH on resting ECG may indicate severe or poorly controlled hypertension and is associated with increased ASCVD risk.⁷⁵ Addition of resting ECG to the standard GRS assessment may be helpful to reclassify ASCVD risk especially in patients with hypertension.^{74,76}

Table 1. 14 Resting Electrocardiogram (ECG)

COR	LOE	Recommendation
IIa	C	A resting ECG can be helpful for ASCVD risk assessment in asymptomatic men and women with hypertension and/or diabetes.
IIb	C	A resting ECG may be helpful for ASCVD risk assessment in asymptomatic men and women without hypertension and diabetes.

1.15. Measurement of Hemoglobin A1c (HbA1c)

HbA_{1c} can be used to diagnose diabetes (HbA_{1c} > 6,5%) and also to identify individuals who are at risk for developing diabetes (HbA_{1c} 5.7% to 6.4%).⁷⁷ Some studies have observed that higher levels of HbA_{1c} are associated with an increased risk of ASCVD in non-diabetic patients.⁷⁸ For non-diabetic individuals, each 1% increase of HbA_{1c} is associated with an adjusted 40% higher risk of ASCVD.⁷⁹ HbA_{1c} is a better predictor to improve risk assessment and reclassification compared to fasting blood glucose.⁸⁰

Table 1. 15 Measurement of Haemoglobin A1c (HbA1c)

COR	LOE	Recommendation
Iib	B	Measurement of HbA1c can be helpful for ASCVD risk assessment in asymptomatic men and women without diabetes.

1.16. Measurement of hemoglobin A1c (HbA1c)

Treadmill test is the most commonly used method for exercise testing and a number of exercise protocols are used during which, the walking speed and the grade are gradually increased in stages. Although the most important abnormality is ST-segment depression during and post exercise, other abnormalities such as exercise capacity, chronotropic response, heart rate recovery and exercise induced arrhythmia are also considered to be associated with increased ASCVD risk.^{81,82} Duke score that incorporates exercise capacity, exercise induced angina and ST segment depression is used to synthesize a score to predict the risk.⁸³

An exercise ECG may be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk, men and women.^{84,85,86} Particular attention should be paid to non-ECG abnormalities such as exercise capacity for risk prediction.^{84,85,86}

Table 1. 16 Performing of Exercise ECG

COR	LOE	Recommendation
Iib	B	Exercise ECG may be helpful for ASCVD risk assessment in asymptomatic, intermediate-risk, men and women.

1.17. Transthoracic Echocardiography

Regional wall motion abnormalities are the most common echocardiographic manifestations of CAD but are only present during or recent (myocardial stunning) of an ischemic event or previous myocardial infarction as evident by loss of muscle mass and/or scar tissue. Additional echocardiographic manifestations of CAD include ischemic mitral regurgitation, reduced left and right ventricular systolic dysfunction. None of the abnormalities mentioned above have satisfactory sensitivity and specificity to consider transthoracic echocardiography for risk assessment of ASCVD.

LVH is associated with increased risk of ASCVD and all-cause mortality independent of blood pressure.^{87,88,89} Echocardiography is more sensitive than resting ECG in diagnosing LVH and also able to measure left ventricular mass index. Patients with uncomplicated essential hypertension has a 40% higher risk of ASCVD for each 39 g/m² greater left ventricular mass index.⁹⁰ Therefore, transthoracic echocardiography to detect LVH for ASCVD risk assessment is only recommended for patients with hypertension.^{90,91}

Table 1. 17 Table Transthoracic Echocardiography

COR	LOE	Recommendation
Iib	B	Transthoracic echocardiography to detect LVH may be helpful for ASCVD risk assessment in asymptomatic men and women with hypertension.
III No Benefit	C	Transthoracic echocardiography is not recommended for ASCVD risk assessment in asymptomatic men and women without hypertension.

1.18. Myocardial perfusion Imaging (MPI)

Exercise or pharmacological stress MPI using single-photon emission computed tomography (SPECT) or positron emission tomography (PET) may be used for ASCVD risk assessment in asymptomatic individuals. However, both SPECT and PET scans involve considerable exposure to ionizing radiation. The average radiation exposure for stress MPI SPECT is about 5 mSv while for stress MPI PET is about 5 mSv after using dose reduction strategies.^{92,93}

There is a well-established correlation between the CCS and the presence of silent ischemia detected by stress MPI.⁹⁴ The presence of silent ischemia is significantly high among patients with higher CCS and diabetes⁹⁵ or higher CCS and individuals with strong family history of premature CAD.⁹⁶

Stress MPI may be helpful in the advanced assessment of asymptomatic individuals who have been evaluated with other risk assessment modalities and found to be at high-risk of silent ischemia. These individuals include patients with diabetes and those with strong family history of premature CAD and having high coronary calcium scores of > 400 AU. Stress MPI is not recommended for ASCVD risk assessment in asymptomatic, low or intermediate-risk, individuals. A special consideration should be made about the amount of radiation exposure when deciding the benefit of stress MPI in an asymptomatic individual against the potential risk.

Table 1. 18 Myocardial perfusion Imaging (MPI)

COR	LOE	Recommendation
Iib	C	Stress MPI may be helpful for ASCVD risk assessment in asymptomatic men and women with diabetes, strong family history of premature ASCVD or other risk assessments suggest high risk of ASCD (e.g., CCS $\geq$ 400 AU).
III No Benefit	C	Stress MPI is not recommended for ASCVD risk assessment in asymptomatic, low or intermediate risk, men and women.

1.19. Obstructive Sleep Apnea (OSA)

OSA affects 25% of men and 9% of women and is associated with increased cardiovascular morbidity and mortality.⁹⁷ Hypertension, myocardial infarction, stroke, heart failure and arrhythmia including atrial fibrillation are associated with OSA.

The increased CVD risk of OSA may be due to increased sympathetic activity, elevated blood pressure and increased oxidative stress secondary to episodic hypoxemia. The oxidative stress increases circulating levels of inflammatory markers causing endothelial dysfunction and atherosclerosis.⁹⁷

Table 1. 19 Obstructive Sleep Apnoea (OSA)

COR	LOE	Recommendation
Iib	C	Assessment for cardiac risk factors (e.g., hypertension, diabetes, hypercholesterolemia, obesity and non-sustained atrial or ventricular arrhythmia) and symptoms and signs (e.g., angina or dyspnea on exertion and poor exercise tolerance) of heart disease can be helpful for CVD risk assessment in men and women with OSA.

1.20. Genotype Testing

Many genetic tests are currently available to identify the genetic risk for CVD. However, there is no agreement regarding which genetic markers to use and how to calculate genetic risk scores. There is no evidence to demonstrate that genotype testing is helpful in decision making to alter management and to improve outcomes in asymptomatic men and women.⁹⁸ Epigenetics involve the chemical alteration of DNA that affect expression of genes. Higher methylation of DNA is involved with higher risk of ASCVD while lower methylation of DNA is associated with higher-risk.⁹⁹ However, there is no evidence to confirm that epigenetic screening is helpful ASCVD risk prediction beyond conventional risk factors.

Table 1. 20 Genotype Testing

COR	LOE	Recommendation
III No Benefit	B	Genotype testing and epigenetic screening for ASCVD risk assessment in asymptomatic men and women are not recommended.

1.21. Measurement of Natriuretic Peptide Levels

Atrial natriuretic peptides, B-type natriuretic peptides and their precursors are measured to diagnose and to assess prognosis of heart failure in symptomatic patients. Increased levels of these markers are also associated with poor prognosis in patients with acute coronary syndrome and stable CAD. However, the value of natriuretic peptides in predicting development of CVD in asymptomatic individuals who are free from CAD and heart failure is not confirmed.¹⁰⁰ Therefore, measurement of natriuretic peptide levels for assessment of CVD risk in asymptomatic individuals is not recommended.

Table 1. 21 Measurement of Natriuretic Peptide Levels

COR	LOE	Recommendation
III No Benefit	B	Measurement of natriuretic peptide levels for CVD risk assessment in asymptomatic men and women is not recommended.

1.22. Measurement of Lipoprotein-Associated Phospholipase A₂ Level

Lipoprotein-associated phospholipase A₂ is an enzyme produced by macrophages and lymphocytes. This is a proatherogenic enzyme that hydrolyses oxidative phospholipids in LDL, generating proatherogenic phospholipids, non-esterified fatty acids and inflammatory markers.¹⁰¹

In a meta-analysis of 14 studies, there was no significant association between lipoprotein-associated phospholipase A₂ enzyme levels and ASCVD risk.¹⁰² Therefore, measurement of lipoprotein-associated phospholipase A₂ level for ASCVD risk assessment in asymptomatic individuals is not recommended.

Table 1. 22 Measurement of Lipoprotein-Associated Phospholipase A₂ Level

COR	LOE	Recommendation
III No Benefit	B	Measurement of lipoprotein-associated phospholipase A ₂ level for ASCVD risk assessment in asymptomatic men and women is not recommended.

1.23. Measurement of Arterial Stiffness

An increase in arterial stiffness is associated with arterial wall damage, especially in hypertensive patients.¹⁰³ Arterial stiffness is usually measured by either aortic pulse wave velocity or arterial augmentation index.¹⁰⁴ However, the use of arterial stiffness for risk prediction of ASCVD is restricted only to research setting.¹⁰⁴ Therefore, systematic use of arterial stiffness for ASCVD risk assessment in general public is not recommended.

Table 1. 23 Measurement of Arterial Stiffness

COR	LOE	Recommendation
III No Benefit	B	Measurement of arterial stiffness for ASCVD risk assessment in asymptomatic men and women is not recommended.

1.24. Computerized Tomography Coronary Angiography (CTCA)

CTCA has been compared with conventional invasive coronary angiography for detection of significant coronary stenosis of > 50% in many studies.¹⁰⁵ Most of these studies show sensitivity and specificity in the range of 85% to 95% and has a high negative predictive value of > 98%.¹⁰⁵

In a study of asymptomatic adults without a history of established CAD, coronary calcium was detected in 18% of participants while identifiable atheromatous plaques were detected in 22% of participants.¹⁰⁶ The prevalence of significant coronary stenosis in low, intermediate and high-risk groups were 2%, 7% and 16% respectively. During the next 18-month follow-up, 15 cardiac events occurred but 14 of these were revascularization procedures based on CTCA results. The utility of CTCA for risk assessment of ASCVD in asymptomatic adults could not be established as there was only one non-procedural cardiac event in this study.

Table 1. 24 Computerized Tomography Coronary Angiography (CTCA)

COR	LOE	Recommendation
III No Benefit	C	CTCA for ASCVD risk assessment in asymptomatic men and women is not recommended.

1.25. Stress Echocardiography

Exercise or pharmacological stress echocardiography is usually performed in symptomatic patients to assist in the diagnosis of obstructive CAD. Stress echocardiography may be considered for testing asymptomatic individuals during pre-operative assessment,¹⁰⁷ for patients with atrial fibrillation and investigation following an episode of ventricular tachycardia or syncope. However, there is little published data on use of stress echocardiography in asymptomatic patients for ASCVD risk assessment.

Table 1. 25 Stress Echocardiography

COR	LOE	Recommendation
III No Benefit	C	Stress echocardiography for CVD risk assessment in asymptomatic, low or intermediate risk, men and women is not recommended.

1.26. Magnetic Resonance Imaging (MRI) of Plaque

MRI is a non-invasive method of assessment of the size and characteristics of atheromatous plaques in aorta, carotid and femoral arteries.^{108,109} Also, MRI does not require exposure to ionizing radiation. The capability of MRI in detection and quantification of atheromatous plaques can be used for risk assessment of ASCVD.¹¹⁰ However, there is no published data to date to confirm the benefit on using MRI plaque assessment in aorta, carotid and femoral arteries for ASCVD risk assessment.

Table 1. 26 Magnetic Resonance Imaging (MRI) of Plaque

COR	LOE	Recommendation
III No Benefit	C	MRI for detection and quantification of atheromatous plaques in aorta, carotid and femoral arteries as ASCVD risk assessment in asymptomatic men and women is not recommended.

1.27. Measurement of Lipoproteins and Apolipoproteins

Additional lipid parameters such as apolipoprotein B (ApoB) is related closely with LDL cholesterol level while apolipoprotein A (ApoA) is related with HDL cholesterol levels. Therefore, measuring these particles beyond standard fasting lipid profile is not beneficial as measurement of non-HDL cholesterol reflects the total concentration of atherogenic peptides and particle number.¹¹¹ Non-HDL cholesterol is a simple measurement, which is calculated as the difference between total and the HDL cholesterol level. There is no publish data to demonstrate an incremental predictive value of LDL subfractions beyond that of conventional CVD risk factors.¹¹²

In the Framingham Heart Study, only a marginal additional risk prediction could be done from ApoB and ApoB/A-1 ratio when compared with total/HDL cholesterol ratio.¹¹³

Lipoprotein (a) [Lp (a)] is a large glycoprotein attached to an LDL like particle and is believed to be associated with increased ASCVD risk that is independent of conventional lipid parameters. A meta-analysis of prospective studies involving 126,634 participants and a follow-up of 1.3 million person-years demonstrated that Lp(a) concentration is weakly correlated with several conventional CVD risk factors.¹¹⁴ There was a continuous, independent but modest association of Lp(a) concentration with risk of ASCVD and stroke. However, the information of independent risk prediction by measuring Lp(a) concentration beyond fasting lipid profile measurement is lacking. Also, there are concerns about measurement and standardization of Lp(a) concentration in a clinical setting.¹¹

Therefore, measurement of apolipoprotein levels, particle size and density and lipoproteins for ASCVD risk assessment in asymptomatic individuals is not recommended.

Table 1. 27 Measurement of Lipoproteins and Apolipoproteins

COR	LOE	Recommendation
III No Benefit	C	Measurement of apolipoprotein levels, particle size and density and lipoproteins for CVD risk assessment in asymptomatic men and women is not recommended.

Chapter 2

Measures for Primordial and Primary Prevention of Cardiovascular Events

2.1. Diet and Nutrition

Specific nutrients in the diet, specific foods and dietary patterns influence ASCVD risk by both their effect on risk factors such as diabetes, hypertension, hypercholesterolemia and obesity and through direct effect.¹¹⁶

Fats and fatty acids

The types of fatty acids consumed are more important than the amount of total fat for ASCVD risk reduction. Intake of saturated fatty acids should be reduced to less than 10% of total energy intake by replacing it with polyunsaturated fatty acids.¹¹⁷ Polyunsaturated fatty acids lower LDL cholesterol and also HDL cholesterol to a lesser extent. Monounsaturated fatty acids increase HDL cholesterol level. There is only weak evidence to confirm that both polyunsaturated and monounsaturated fatty acids lower ASCVD risk.¹¹⁸

Trans fatty acids are a subclass of saturated fatty acids that forms during industrial processing of fats. Trans fats increase total cholesterol and reduce HDL cholesterol level. Increasing the energy intake from trans fats by 2% leads to increase ASCVD risk by 23%.¹¹⁹ It is recommended to reduce trans fats to < 1% of total energy intake.

The impact of the amount of dietary cholesterol to serum cholesterol is weak compared to composition of fatty acids in the diet. However, it is recommended to reduce cholesterol intake to < 300 mg per day.

Minerals

The Dietary Approaches to Stop Hypertension (DASH) trial demonstrated the association between sodium restriction and blood pressure reduction.¹²⁰ Almost 80% of the total salt comes from processed food while remaining is added during cooking or at the dining table. It is recommended to limit salt intake to < 5 g per day.

Increasing potassium intake contributes to lower blood pressure.¹²¹ Higher consumption of fruits and vegetables, which are high in potassium should be considered in addition to salt restriction to control high blood pressure.

Vitamins

Serum levels of vitamin A, B, C and E have been shown no association between these vitamins and ASCVD risk. Vitamin D₃ level has shown a small beneficial effect by reducing all-cause mortality by 11%.¹²² However, it is not clear whether Vitamin D₃ specifically reduces CVD related mortality. Therefore, Vitamin D₃ supplements for ASCVD risk reduction is not recommended.

Fiber

Consuming > 7 g per day of fiber from whole grain products is associated with 9% reduction of ASCVD risk.¹²³

Nuts

Daily consumption of 30 g of nuts reduces ASCVD risk by about 30%.¹²⁴

Fruits and vegetables

Consumption of each additional serving (a palm full) of fruits and vegetables up to 5 sessions per day reduces the ASCVD related mortality by 4%.¹²⁵ Therefore, 5 servings of fruits or vegetables are recommended for ASCVD risk reduction.

Fish and fish oils

Eating fish at least once a week is associated with 16% reduction of ASCVD risk compared with eating less fish¹²⁶ and eating fish 2-4 times a week reduces the risk of stroke by 6% compared with eating fish less than once a week.¹²⁷

However, using fish oil supplements have no benefit in reducing ASCVD risk or cardiac events.¹²⁸

Alcohol

Alcohol is a colossal global health issue and small reductions in health-related harms at low levels of alcohol intake are outweighed by the increased risk of other health-related harms, including cancer.¹²⁹

Sugar and sugar-sweetened soft drinks

Regular consumption of sugar and sugar-sweetened soft drinks is associated with obesity, diabetes and metabolic syndrome. The WHO recommends to limit the energy from sugar and sugar-sweetened soft drinks to < 10% of daily total energy intake.¹³⁰

Table 2. 1 Sugar and sugar – sweetened soft drinks

COR	LOE	Recommendation
I	B	A diet containing higher amounts of vegetables, fruits, legumes, nuts, whole grains and fish is recommended for ASCVD risk reduction.
IIa	B	Replacement of saturated fat in the diet with monounsaturated and polyunsaturated fats can be helpful for ASCVD risk reduction.
IIa	B	Cholesterol and salt restriction in the diet can be helpful for ASCVD risk reduction.
IIa	B	Minimizing the intake of processed meats, refined carbohydrates and sweetened beverages can be helpful for ASCVD risk reduction.
III No Benefit	B	Using fish oil supplements are not recommended for ASCVD risk reduction.
III Harm	B	Trans fats should be avoided for ASCVD risk reduction.

2.2. Physical Activity and Exercises

Regular physical activities and exercises help to prevent and control many cardiovascular risk factors including hypertension, diabetes, LDL cholesterol and non-HDL cholesterol.¹³¹ Regular physical activities and exercises also reduce CVD related mortality and all-cause mortality in healthy individuals and patients with cardiovascular risk factors.¹³²

Aerobic physical activities and exercises have a dose-response effect on reduction of ASCVD risk and cardiac events.^{133,134,135} These exercises consist of movement of large muscles in a rhythmic manner for a period of time. They include brisk walking, jogging, swimming, cycling, dancing, gardening and daily household and occupational activities.

Aerobic physical activities and exercises of moderate intensity for at least 150 minutes or vigorous intensity for at least 75 minutes per week is recommended for all healthy men and women for ASCVD risk reduction.^{133,134,136} The frequency of each session should be at least 5 times a week but preferably every day.

The prescription for physical activity and exercises should be tailored to individuals to determine intensity, duration and frequency accordingly. Individuals who can do more exercises should be encouraged to do more exercises and should be encouraged to do up to 300 minutes of moderate intensity or 150 minutes of vigorous intensity exercises per week to gain additional benefits.^{133,134} Similarly, individuals, who cannot perform minimum recommended level of physical activity and exercises should be encouraged to do some moderate or vigorous activities as even less than minimum recommended level of activities can be beneficial to reduce ASCVD risk.^{137,138}

Sedentary individuals, especially those with high ASCVD risk should undergo a thorough clinical evaluation including exercise ECG before prescribing them physical activities and exercises.¹³⁹

Each session of physical activity or exercises should include warm-up, conditioning and cool-down phases. Warming-up prior to exercises and cooling-down following exercises with stretching and flexing can be helpful to prevent injuries and adverse cardiac events.

Other types of exercises such as dynamic resistance exercises and isometric resistance exercises are associated with enhancing muscle mass, strength and power and exercise capacity.¹⁴⁰ Neuromotor exercises are also helpful to maintain and improve balance, coordination, agility and gait, especially in elderly individuals. However, the beneficial effects in terms of ASCVD risk reduction by these exercises have not been demonstrated.

Table 2. 2 Physical Activity and Exercises

COR	LOE	Recommendation
I	A	At least 150 minutes of moderate intensity or 75 minutes of vigorous intensity aerobic physical activity a week is recommended for healthy men and women of all ages for ASCVD risk prevention.
I	A	Gradual increase of aerobic physical activity to 300 minutes of moderate intensity or 150 minutes of vigorous intensity a week is recommended for healthy men and women for additional benefits in ASCVD risk prevention.
IIa	B	For men and women who are unable to meet minimum recommendations of physical activity, engaging some moderate or vigorous intensity physical activity, even if less than the recommended amount, can be helpful to reduce ASCVD risk.
IIa	C	Clinical evaluation, including PAR-Q screening and exercise ECG, should be considered for sedentary men and women with cardiovascular risk factors and intending to start vigorous physical activity or sports.

Whether to change the whole chart accordingly or do we need to have this separately as a new document.

2.3. Tobacco cessation

Tobacco affects vascular endothelial inflammation, oxidative processes, platelet functions, fibrinolytic mechanisms, lipid oxidation and vasomotor functions. These mechanisms enhance atherosclerotic process and prothrombotic phenomena increasing the risk of ASCVD and cardiac events.

Smoking cessation is the most cost-effective intervention for ASCVD risk reduction. The risk associated with smoking is related to the degree of smoking with no lower limit for its deleterious effects.¹⁴¹ A long-standing smoker has a 50% risk of dying from his smoking habit and average loss of 10 years of his life.¹⁴² The 10-year risk of mortality is approximately double in smokers compared to non-smokers. Even low-level smoking is associated with increased ASCVD risk.¹⁴³

Passive smoking or environmental tobacco smoking is also associated with increased ASCVD risk.^{144,145} The risk of ASCVD increases by 30% if an individual is exposed to passive smoking.

The risk of ASCVD is reduced almost close to the level of a non-smoker within 10-15 years of abstinence from smoking. There is no age limit to achieve the benefit of smoking cessation. Passive smoking should also be avoided.

Repeated advice, counselling including motivational interviews and support are important to help smokers to stop smoking. The pharmacological therapy to help smoking cessation includes nicotine replacement therapy (NRT), varenicline and bupropion.¹⁴⁶ In Sri Lanka consensus was agreed among the stakeholders i.e Ceylon college of Psychiatrist, NATA, Directorate of Mental Health not to recommend NRT based on evidence.¹⁴⁷

Electronic cigarettes (e-cigarettes) are battery operated devices that vaporize nicotine and other chemicals to inhale and obtain the effects of combustible cigarettes. Studies have shown that the effects of e-cigarettes are similar to NRT trans dermal patches¹⁴⁸ and inhalers.¹⁴⁹ More research is required to study the long-term safety and health effects of e-cigarrets.¹⁵⁰ According to the National Authority on Tobacco and Alcohol Act, no 27 of 2006 gazette, dated 31st August 2016 e cigarettes are banned in Sri Lanka.

Table 2. 3 Tobacco Cessation

COR	LOE	Recommendation
I	A	All men and women should be advised to abstain from smoking tobacco, other herbal products and consuming smokeless tobacco products.
I	A	Repeated advice, counselling, support and medications including varenicline and bupropion individually or in combination are recommended for all smokers and people who use smokeless tobacco products to help tobacco cessation.
I	B	All men and women should be advised to avoid exposure to passive smoking (environmental tobacco smoking).

2.4. Weight Control

Overweight and obesity are associated with increase in blood pressure, insulin resistance, dyslipidemia, albuminuria and prothrombotic state. Both overweight and obesity are associated with development of diabetes, CAD, heart failure, atrial fibrillation, stroke, CVD related mortality and all-cause mortality. Therefore, weight loss is recommended for all overweight and obese individuals for ASCVD risk reduction.¹⁵¹

Body mass index (BMI) [weight (kg)/ (height (m))²] is used extensively to identify overweight and obese men and women. Individuals with BMI of 25-29.9kg/m² are categorized as overweight while BMI > 30 kg/m² are categorized as obese. All-cause mortality is lowest with BMI of 20-25 kg/m². However, maintaining a BMI between 20-23 kg /m² is recommended for Sri Lankan population considering the higher risk of ASCVD among South Asians.

BMI 23-24.9kg/m² is considered as an increased risk for overweight. (Therefore, it is considered as a trigger point for life style modification)

The pattern of fat distribution is especially important for South Asians as their intra-abdominal fat deposition is associated with higher cardiometabolic and ASCVD risks. Waist circumference and waist to hip ratio (WHR) are used to identify individuals with abdominal obesity.¹⁵² For South Asian populations, men with waist circumference $\geq$ 90 cm and WHR > 0.9 and women with waist circumference $\geq$ 80 cm and WHR > 0.8 are recommended weight loss to reduce cardiometabolic and ASCVD risk.¹⁵³

Life style interventions, behavioral modifications, exercises and calorie restriction are important for weight loss.^{151,154} Medical therapy for weight loss include orlistat and bariatric surgery. Patients who undergo bariatric surgery have a reduced risk of CVD events including myocardial infarction and stroke and mortality compared with non-surgical controls.¹⁵⁵

Table 2. 4 Weight control

COR	LOE	Recommendation
I	B	Weight loss is recommended 0.5 kg/week until optimal BMI is reached to all men and women with overweight and obesity for ASCVD risk reduction.
I	B	Lifestyle interventions and counselling are recommended for men and women with overweight and obesity to achieve and maintain weight loss.
I	C	Measurement of BMI is recommended for all men and women to identify overweight and obesity for weight loss considerations.
IIa	B	Measurement of waist circumference can be helpful in identifying men and women with higher cardiometabolic risk.

2.5. Management of Diabetes

Patients with diabetes has double the risk of ASCVD when compared to non-diabetic patients.¹⁵⁶ Lifestyle modifications including heart-healthy diet for optimal glycemic control, weight management, regular exercises and physical activity and smoking cessation are of paramount importance for the management of diabetes.¹⁵⁷ The target HbA1c should be maintained below 7% in patients with both Type I and Type II diabetes.^{158,159} In addition, optimal blood pressure control and lipid lowering therapy also play an important role in ASCVD risk reduction in diabetic patients.

The dietary plan of diabetic patients should include predominance of fruits, vegetables, green leaves, whole grain cereals and low-fat proteins while high-carbohydrate diets, sugar, sugar-sweetened drinks, salt, saturated fats, Trans fats and alcohol should be restricted.

At least 150 minutes per week of moderate-intensity exercises or 75 minutes per week high-intensity exercises are recommended for optimal glycemic control and ASCVD risk reduction. These exercises can be a combination of aerobic and resistance exercises.

Smoking should be strongly discouraged as this is associated with ASCVD and premature deaths among both diabetic and non-diabetic patients.^{157,160}

Although short duration of diabetes does not increase the ASCVD risk significantly, the risk approaches to CAD risk equivalence after about a decade and in those with proteinuria and reduced eGFR.^{161,162,163}

Meta-analyses involving data from multiple, large, randomized, controlled studies have shown that intense glucose control significantly reduces non-fatal myocardial infarction and CAD events but no effect on stroke or all-cause mortality.^{164,165} Metformin is the first-line treatment recommended for glycemic control if this medication is tolerated and not contraindicated.¹⁶⁶ In a large, randomized, controlled study, metformin reduced microvascular and macrovascular complications by 32%, myocardial infarctions by 35% and all-cause mortality by 36% when compared with life-style modifications alone.¹⁶⁷

Newer hypoglycemic agents, SGLT2 inhibitors and GLP-1R agonists may be effective in CVD risk reduction in diabetic patients. Addition of SGLT2 inhibitor, empagliflozin, to diabetes management was associated with reduction of CVD by 38%, hospitalization for heart failure by 35% and all-cause mortality by 35% when compared to patients with standard diabetic management without a SGLT2 inhibitor.¹⁶⁸ However, non-fatal myocardial infarctions and strokes were not reduced by empagliflozin. These results suggest that the mechanism of benefit was likely to be due to cardio-renal hemodynamic effects than hypoglycemic action or prevention of atherosclerosis. Two other randomized, controlled trials also showed significant reduction of CVD events and heart failure by other types of SGLT2 inhibitors, canagliflozin¹⁶⁹ and dapagliflozin.¹⁷⁰

GLP-1R agonists also have been shown to reduce ASCVD risk significantly. These effects have been demonstrated with liraglutide,¹⁷¹ albiglutide¹⁷² and semaglutide.¹⁷³

There is clear evidence to confirm that intense lipid lowering therapy using simvastatin¹⁷⁴ or atorvastatin¹⁷⁵ is associated with a significant reduction of ASCVD events and related mortality in

diabetic patients who have no history of CAD or myocardial infarction. There is greater reduction of ASCVD risk with more intense lipid lowering therapy in diabetic patients.¹⁷⁶ Also, addition of ezetimibe to statin therapy for more intense LDL cholesterol control is associated with further reduction of ASCVD risk.¹⁷⁷ Lipid lowering therapy is recommended for all diabetic patients > 40 years of age with high-risk patients should achieve LDL cholesterol target of < 70 mg/dL (non-HDL cholesterol target < 100 mg/dL) while intermediate-risk patients should have a LDL cholesterol target of < 100 mg/dL (non-HDL cholesterol target < 130 mg/dL).

Optimal blood pressure control is also important alongside with glycemic control and lipid lowering therapy in diabetic patients. A systematic review and a meta-analysis of randomized trials of blood pressure lowering in > 100,000 diabetic patients confirmed that optimal blood pressure control reduces all-cause mortality, CVD events, strokes and heart failure in these patients.¹⁷⁸ Angiotensin converting enzyme inhibitors (ACE-I) or angiotensin receptor blockers (ARBs) are the first line anti-hypertensive therapy in diabetic patients as these medications are associated with initiation and prevention of diabetic nephropathy. A target blood pressure of < 130/80 mmHg should be achieved in both Type I and Type II diabetic patients to reduce ASCVD risk and diabetic nephropathy.

Low-dose aspirin is generally not indicated for primary prevention of ASCVD in diabetic patients. However, low-dose aspirin (75-100 mg daily) may be considered for diabetic patients with high-risk for ASCVD (See 5.10).

Intense glycemic control, lipid lowering therapy and blood pressure control are important for ASCVD risk reduction in type I diabetic patients as well. A Swedish study has demonstrated that type I diabetic patients with HbA_{1c} < 6.9% have 2 times higher risk while patients with HbA_{1c} > 9.7% has about 10 times higher risk of ASCVD when compared to general population.¹⁷⁹ Same HbA_{1c}, blood pressure and LDL cholesterol targets for type II diabetic patients are recommended for type I diabetic patients as well.

Table 2. 5 Management of Diabetes

COR	LOE	Recommendation
I	A	A heart-healthy dietary pattern is recommended for ASCVD risk reduction in all men and women with diabetes aiming at optimal glycemic control and weight maintenance.
I	A	At least 150 minutes per week of moderate-intensity exercises or 75 minutes per week of high-intensity exercises are recommended for ASCVD risk reduction in all men and women with diabetes.
I	A	A target HbA1c < 7% is recommended for reduction of ASCVD risk and microvascular complications in all men and non-pregnant women with both Type I and Type II diabetes.
I	A	Lipid lowering therapy, preferably with statins, is recommended for ASCVD risk reduction in all Type I and Type II diabetic patients above the age of 40 years. Diabetic patients with high-risk (10-year risk of fatal or non-fatal CVD event $\geq 20\%$) of ASCVD should have an LDL cholesterol target of < 70 mg/dL (non-HDL cholesterol target < 100 mg/dL) while intermediate-risk (10-year risk of fatal or non-fatal CVD event between 10- <20%) patients should have an LDL cholesterol target of < 100 mg/dL (non-HDL cholesterol target < 130 mg/dL).
I	B	Lowering of blood pressure to < 130/80 mmHg is recommended for patients with both Type I and Type II diabetes and hypertension.
IIa	B	Metformin can be helpful as the first-line drug therapy alongside with life-style modifications for optimal glycemic control and ASCVD risk reduction in men and women with type II diabetes.
IIb	A	Lipid lowering therapy, preferably with statins, may be helpful for ASCVD risk reduction in Type I and Type II diabetic patients below the age of 40 years but significantly elevated risk, based on the presence of albuminuria or multiple cardiovascular risk factors. Diabetic patients with high-risk of ASCVD should have an LDL cholesterol target of < 70 mg/dL (non-HDL cholesterol target < 100 mg/dL) while intermediate risk patients should have an LDL cholesterol target of < 100 mg/dL (non-HDL cholesterol target < 130 mg/dL).
IIb	B	SGLT-2 inhibitors or GLP-1R agonists may be helpful to achieve glycemic control and CVD risk reduction in type II diabetic patients when their life-style modifications and metformin therapy are insufficient for optimal glycemic control.

2.6. Management of Hypertension

Hypertension is defined as an office measurement of systolic blood pressure (SBP) ≥ 140 mmHg and/or diastolic blood pressure (DBP) ≥ 90 mmHg. Hypertension is a major risk factor for CAD, stroke, heart failure, CKD, atrial fibrillation and peripheral vascular disease. There is a log-linear relationship of mortality from CAD and stroke with increasing SBP from 115 mmHg and DBP from 75 mmHg upwards.¹⁸⁰

The decision to start anti-hypertensive therapy is based on the measured blood pressure and total CVD risk of the patient. Lifestyle modifications including weight reduction, regular exercises,¹⁸¹ salt restriction,¹⁸² alcohol moderation and a dietary pattern¹⁸³ with increased consumption of fruits, vegetables and low-fat dairy products are recommended for all men and women with hypertension and borderline hypertension.

Anti-hypertensive therapy is recommended for patients with grade I hypertension (SBP 140-159 mmHg and/or DBP 90-99 mmHg) and grade II hypertension (SBP 160-179 mmHg and/or DBP 100-109 mmHg) if their blood pressure remain in this range during several office and/or ambulatory blood pressure measurements despite reasonable life-style modifications.¹⁸⁴

Prompt commencement of anti-hypertensive therapy is recommended for patients with grade III hypertension (SBP ≥ 180 mmHg and/or DBP ≥ 110 mmHg) irrespective of their CVD risk.¹⁸⁵

All major antihypertensive drug classes including ACE-I, ARBs, calcium channel blockers, diuretics and beta blockers are effective in lowering blood pressure and to reduce CVD morbidity and mortality when they are used either as monotherapy or in combination.^{185,186} These benefits are mainly due to lowering of blood pressure per say independent of the class effect of the anti-hypertensive medication used.^{185,186}

Anti-hypertensive medications with 24-hour efficacy improves compliance and reduce blood pressure variability, organ damage and CVD risk. ACE-I and ARBs should be used with caution in women of child bearing age due to potential teratogenic effects of these medications.¹⁸⁷

Combinations of anti-hypertensive medications may be needed to achieve target blood pressure control in many patients. The additional blood pressure reduction by combining two anti-hypertensive medications from two different classes is approximately 5 times more than doubling the dose of one medication¹⁸⁸ and cause less side effects.

Reduction of blood pressure to a target of < 140 mmHg of SBP and < 90 mmHg of DBP is recommended for all hypertensive patients < 60 years of age.¹⁸⁹ Reduction of SBP to a target between 140 to 150 mmHg is recommended for all hypertensive patients > 60 years of age and > 80 years with good physical and mental conditions when they are found to have a SBP > 160 mmHg.¹⁹⁰

Reduction of blood pressure to a target of < 130 mmHg of SBP and < 80 mmHg of DBP is recommended for all hypertensive patients with CKD^{191,192,193} and diabetes.¹⁹²

Table 2. 6 Management of Hypertension

COR	LOE	Recommendation
I	A	Lifestyle modifications including weight reduction, regular exercises, salt restriction, and increased consumption of fruits, vegetables and low-fat dairy products are recommended for all men and women with hypertension and borderline hypertension.
I	A	All major antihypertensive drug classes including ACE-I, ARBs, calcium channel blockers, diuretics and beta blockers are recommended for blood pressure lowering as they do not differ significantly in their blood pressure lowering efficacy.
I	B	Reduction of blood pressure to a target of < 140 mmHg of SBP and < 90 mmHg of DBP is recommended for all the hypertensive men and women < 60 years of age.
I	B	Reduction of blood pressure to a target SBP between 140 to 150 mmHg is recommended for all the hypertensive men and women > 60 years of age and SBP > 160 mmHg.
I	B	Reduction of blood pressure to a target SBP between 140 to 150 mmHg is recommended for all the hypertensive men and women > 80 years of age and SBP > 160 mmHg provided they are in good physical and mental conditions.
I	B	Reduction of blood pressure to a target of < 130 mmHg of SBP and < 80 mmHg of DBP is recommended for all the hypertensive men and women with CKD.
I	B	Reduction of blood pressure to a target of < 130 mmHg of SBP and < 80 mmHg of DBP is recommended for all the hypertensive men and women with diabetes.

2.7. Management of Dyslipidemia

There is a strong association between total cholesterol as well as LDL cholesterol levels and ASCVD risk.¹⁹⁴ This association applies to men and women with and without CAD. Meta-analysis of several trials of lipid lowering therapy with statins clearly show a linear relationship of lipid lowering and reduction of ASCVD. The ASCVD mortality and non-fatal myocardial infarctions are reduced by 20-25% with each 40 mg/dL (1.0 mmol/L) reduction of LDL cholesterol.¹⁹⁵

Non-HDL cholesterol comprises of all pro-atherogenic lipoproteins including LDL cholesterol, intermediate-density lipoprotein cholesterol and very low-density lipoprotein cholesterol. This can be calculated easily by subtracting HDL cholesterol from total cholesterol level. It is a better prediction of ASCVD than LDL cholesterol.¹⁹⁶ non-HDL cholesterol is a more accurate measurement than calculated LDL cholesterol values when triglyceride levels > 400 mg/dL (> 4.5 mmol/L) and also has the advantage of not requiring patients to fast before giving the blood sample.

Low HDL cholesterol is associated with increased ASCVD risk.¹⁹⁷ HDL cholesterol < 40 mg/dL (< 1.0 mmol/L) in men and < 45 mg/dL (< 1.2 mmol/L) in women are regarded as levels of increased risk

for ASCVD. Lifestyle interventions, especially regular exercises and physical activity, are more important than drug therapy to elevate HDL cholesterol levels.

Elevated triglyceride also has an independent but weak association with ASCVD risk.¹⁹⁸ Combination of elevated triglycerides and low HDL cholesterol levels are associated with physical inactivity, abdominal obesity, insulin resistance and type II diabetes. Although, there are no randomized trials to decide on a target level to lower serum triglyceride, fasting triglyceride > 150 mg/dL (> 1.0 mmol/L) is considered as a marker for increased ASCVD risk.

Statins are the first line medications to treat hypercholesterolemia and combined hyperlipidemia. Cholesterol lowering with statin therapy is associated with significantly reduced risk of needing coronary revascularization and ASCVD related morbidity and mortality.^{199,200} In high-risk patients, addition of ezetimibe to statins may provide an additional benefit in ASCVD risk reduction when the highest recommended or maximum tolerated dose of statins is inadequate to lower the LDL cholesterol to the target limits.¹⁷⁷ Addition of fenofibrates to statins are also helpful in lowering triglycerides, increasing HDL cholesterol and reducing LDL cholesterol levels. However, there is no convincing evidence to show that combination of statins and fenofibrate reduce ASCVD risk.

Most patients of 40-75 years with type II diabetes are either intermediate or high-risk for ASCVD. Large RCTs on primary prevention of ASCVD in diabetic patients show a significant risk reduction of ASCVD by lipid lowering using statin therapy.^{201,202,203} Therefore, moderate-intensity statin therapy to reduce LDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 100-200 mg/dL (2.6-5.1 mmol/L) is recommended for reduction of ASCVD risk in all diabetic men and women of age 40-75 years irrespective of their estimated cardiovascular risk.²⁰⁴ Similarly, moderate-intensity statin therapy to reduce non-HDL cholesterol to < 130 mg/dL (< 3.4 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if non-HDL cholesterol is between 130-230 mg/dL (3.4-6.0 mmol/L) is recommended for reduction of ASCVD risk in all diabetic men and women of age 40-75 years irrespective of their estimated cardiovascular risk.

The risk of ASCVD events go even higher when diabetic patients develop additional risk modifiers including long duration of diabetes (type I diabetes > 10 years or type II diabetes > 20 years), microalbuminuria, eGFR < 60 mL/min/1.73m², retinopathy, neuropathy or ABI < 0.9. Therefore, high-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 70-135 mg/dL (1.8-3.5 mmol/L) is recommended for reduction of ASCVD risk in all men and women with multiple CVD risk factors.^{204,205,206,207} Similarly, high-intensity statin therapy to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if non-HDL cholesterol is between 100-165 mg/dL (2.6-4.3 mmol/L) is recommended for reduction of ASCVD risk in all men and women with multiple CVD risk factors.

Individuals who have severe primary hypercholesterolemia with LDL cholesterol $\geq$ 190 mg/dL ($\geq$ 4.9 mmol/L) is associated with high ASCVD risk²⁰⁸ with premature and recurrent coronary events. Because high-intensity statin therapy is associated with greater reduction of ASCVD risk,²⁰⁵ highest recommended or maximum tolerable statin dose should be recommended for these individuals. High-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) is recommended for reduction of ASCVD risk in all men and women of age 20-75 years if their

baseline LDL cholesterol is ≥ 190 mg/dL (≥ 4.9 mmol/L).²⁰⁴ Similarly, high-intensity statin therapy to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) is recommended for reduction of ASCVD risk in all men and women of age 20-75 years if their baseline non-HDL cholesterol is > 220 mg/dL (> 5.7 mmol/L).

The benefit of statin therapy depends on the initial global risk of ASCVD and the intensity of the treatment.^{194,195} High-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 70-135 mg/dL (1.8-3.5 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is $> 20\%$. Similarly, high-intensity statin therapy to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if non-HDL cholesterol is between 100-165 mg/dL (2.6-4.3 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is $> 20\%$.

Moderate-intensity statin therapy to reduce LDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 100-200 mg/dL (2.6-5.1 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is between 10 - $<20\%$. Similarly, moderate-intensity statin therapy to reduce non-HDL cholesterol to < 130 mg/dL (< 3.4 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if non-HDL cholesterol is between 130-230 mg/dL (3.4-6.0 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is between 10 - $<20\%$. Individuals with intermediate-risk according to GRS may be reclassified in to high-risk considering their concomitant risk factors including coronary calcium score ≥ 100 AU,²⁰⁹ inflammatory disease (e.g., rheumatoid arthritis, psoriasis²¹⁰), history of cancer chemotherapy and human immunodeficiency virus infection when treated with protease inhibitors.²¹¹

Table 2. 7 Management of Dyslipidaemia

COR	LOE	Recommendation
I	A	Moderate-intensity statin therapy to reduce LDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) and/or to reduce non-HDL cholesterol to < 130 mg/dL (< 3.4 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 100-200 mg/dL (2.6-5.1 mmol/L) and/or non-HDL cholesterol is between 130-230 mg/dL (3.4-6.0 mmol/L) is recommended for reduction of ASCVD risk in all diabetic men and women of age 40-75 years irrespective of their estimated cardiovascular risk.
I	B	High-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) and/or to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) is recommended for reduction of ASCVD risk in all men and women of age 20-75 years if their baseline LDL cholesterol is > 190 mg/dL (> 4.9 mmol/L) and/or non-HDL cholesterol is > 220 mg/dL (> 5.7 mmol/L). This is for men and women with familial hypercholesterolemia.
I	B	High-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) and/or to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 70-135 mg/dL (1.8-3.5 mmol/L) and/or non-HDL cholesterol is between 100-165 mg/dL (2.6-4.3 mmol/L) is recommended for reduction of ASCVD risk in all men and women with multiple CVD risk factors including long duration (type I diabetes > 10 years or type II diabetes > 20 years), microalbuminuria, eGFR < 60 mL/min/1.73m ² , retinopathy, neuropathy or ABI < 0.9.
I	B	High-intensity statin therapy to reduce LDL cholesterol to < 70 mg/dL (< 1.8 mmol/L) and/or to reduce non-HDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 70-135 mg/dL (1.8-3.5 mmol/L) and/or non-HDL cholesterol is between 100-165 mg/dL (2.6-4.3 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is ≥ 20%.
I	B	Moderate-intensity statin therapy to reduce LDL cholesterol to < 100 mg/dL (< 2.6 mmol/L) and/or to reduce non-HDL cholesterol to < 130 mg/dL (< 3.4 mmol/L) or at least 50% reduction of the respective baseline cholesterol levels if LDL cholesterol is between 100-200 mg/dL (2.6-5.1 mmol/L) and/or non-HDL cholesterol is between 130-230 mg/dL (3.4-6.0 mmol/L) is recommended for reduction of ASCVD risk in all men and women with 10-year estimated GRS for cardiovascular events is between 10 - <20%.

2.8. Behavioral Change

Life-styles of individuals are usually based on their behavioral patterns and the social environment they live. Individualized counselling is important for motivation and commitment to change life-style. Cognitive behavioral interventions such as “motivational interviewing” increase motivation and commitment.²¹² Changing the life-style by a small but a steady step at a time is key to change long-term behaviour.²¹³

Combining the expertise of a multi-disciplinary team including physicians, nurses, dieticians, physical training instructors, psychologists and social workers into a multi-model behavioral intervention can tremendously improve the preventative efforts of ASCVD. Multi-model behavioral interventions include health education, smoking cessation, healthy diet, regular exercises, weight reduction and stress management and counselling for psychosocial risk factors. These types of interventions are especially important in high-risk individuals of ASCVD.

Table 2. 8 Behavioural Change

COR		LOE	Recommendation
I		A	Cognitive behavioral strategies such as “motivational interviewing” are recommended for life-style modifications to reduce ASCVD risk.
I		A	Approach by multidisciplinary health care professionals such as physicians, nurses, dietitians, physical training instructors, psychologists and social workers is important for life-style and behavioral modifications to reduce ASCVD risk.
I		A	Multi model interventions including health education, smoking cessation, healthy diet, regular exercises, weight reduction and stress management and counselling for psychosocial risk factors are recommended for ASCVD risk reduction in high-risk individuals.

2.9. Management of Psychosocial Factors

Individuals with psychological and psychiatric issues such as stress, anxiety, depression, social isolation and negative emotions need special efforts and interventions by psychologists and psychiatrists as these factors act as barriers against their behavioral change.^{214,215} Females, older persons and individuals of low socio-economic status also need programs in order to meet their special needs.

Table 2. 9 Management of Psychosocial Factors

COR	LOE	Recommendation
IIa	A	Referral for psychiatric assessment, psychotherapy, medications and multi-disciplinary management can be helpful for ASCVD risk reduction in patients with depression, anxiety and other psychiatric conditions.

2.10. Low-Dose Aspirin Therapy

The benefit of aspirin in secondary prevention of ASCVD and stroke is well established.²¹⁶ However, the use of aspirin in primary prevention of ASCVD is controversial as it is difficult to balance the benefits and the potential harm. Also, the value of aspirin in primary prevention of ASCVD is becoming less important with the improvements of the co-administration of other primary preventative methods in hypertension, lipid lowering and diabetes management therapies.^{217,218,219}

In terms of primary prevention of ASCVD, aspirin therapy may be helpful only in high-risk individuals with age 40-70 years.^{218,221,222} They include individuals with strong family history of premature myocardial infarction, high coronary artery calcium score,²²² diabetes for > 10 years, micro albuminuria, CKD and inability to achieve target lipid, blood pressure and blood glucose control.²²³ Their bleeding risk should also be assessed before commencing aspirin therapy.

Prophylactic use of aspirin is not recommended for primary prevention of ASCVD in individuals > 70 years because of high-risk of bleeding in this age group outweigh the benefits.²²⁴ Also, aspirin is not recommended for primary prevention of ASCVD in individuals < 40 years of age as there is insufficient evidence to justify the benefits of aspirin over potential risks of this therapy. Although, routine use is not recommended, aspirin may be given to selected high-risk patients in these age groups after discussing the risks and benefits with the patients. The eligible high-risk patients are as mentioned above. Their bleeding risk should also be assessed before commencing aspirin therapy.

Low-dose aspirin is not recommended as a primary preventative measure in ASCVD risk reduction in individuals who have a high-risk of bleeding.²²⁵ Individuals with high-risk of bleeding include those with a history of gastrointestinal bleeding and peptic ulcers, bleeding from other sites, age > 70 years, thrombocytopenia, coagulopathy, CKD and concomitant use of other medications that increases bleeding risks. The medications that increase bleeding risk include non-steroidal anti-inflammatory drugs, steroids, warfarin and direct oral anticoagulants.

Table 2. 10 Low-Dose Aspirin Therapy

COR	LOE	Recommendation
Iib	A	Low-dose aspirin (75-100 mg daily) may be helpful for primary prevention of ASCVD in high-risk patients (e.g., individuals with strong family history of premature myocardial infarction, high coronary artery calcium score, diabetes for > 10 years, micro albuminuria, CKD and inability to achieve target lipid, blood pressure and blood glucose control) of age 40-70 years.
III Harm	B	Low-dose aspirin (75-100 mg daily) is not recommended routinely for primary prevention of ASCVD in men and women < 40 years and > 70 years of age.
III Harm	B	Low-dose aspirin (75-100 mg daily) is not recommended for primary prevention of ASCVD in individuals of any age with increased risk of bleeding.

2.11. Prevention of Heart Disease in Patients Treated for Cancer

Surviving cancer patients, who had been treated with chemotherapy or radiotherapy, have an increased risk of CVD. The increased risk of CVD also depends on the type and the dose of treatment given. The risk further increases when these patients also have other risk factors for CVD.²²⁶

There are 2 types of chemotherapy-associated cardiotoxicity. Type I cardiotoxicity occurs with agents causing irreversible cardiotoxicity (e.g., anthracyclines) while type II cardiotoxicity occurs with agents causing partially reversible cardiotoxicity (e.g., trastuzumab).²²⁷

Both chemotherapy²²⁷ and chest radiotherapy²²⁸ are associated with accelerated atherosclerosis and increased risk of ASCVD. Type I cardiotoxicity mainly causes left ventricular dysfunction. Pretreatment assessment for left ventricular is important²²⁹ and clinical features should be monitored during the treatment for early detection of left ventricular dysfunction in patients receiving potentially cardiotoxic chemotherapy.²³⁰ Beta blockers, angiotensin converting enzyme (ACE) inhibitors and statins have been tested and complied to prevent chemotherapy induced left ventricular dysfunction.²³¹ Early preventative treatment is paramount to exert the maximum benefit.²²⁹

Table 2. 11 Prevention of Heart Disease in Patients Treated for Cancer

COR	LOE	Recommendation
IIa	B	Prophylactic treatment with beta blockers, ACE inhibitors and statins can be helpful for prevention of left ventricular dysfunction in patients receiving high cumulative doses of type I chemotherapy.
IIa	C	Optimization of CVD risk profile can be helpful for minimizing the CVD risk in cancer patients who are receiving chemotherapy or chest radiotherapy.

Chapter 3

Gaps in Evidence

1. There are no recent RCTs for a global risk score assessment approach.
2. Young individuals (< 40 years), elderly (> 75 years) and women are underrepresented in clinical trials.
3. The association between routine screenings for psychosocial risk factors with future ASCVD events needs further evaluation.
4. The role of biomarkers such as hsCRP, natriuretic peptides in risk prediction of ASCVD needs further research and evaluation.
5. ASCVD risk reduction in patients treated with lipid or blood pressure lowering medications as a result of reclassification by CCS or ABI measurements are needed to be proven.
6. No data available on age to commence ASCVD risk assessment and the interval of repeating the assessment.
7. Inadequate data to confirm that pre-eclampsia, pregnancy-induced hypertension and polycystic ovarian disease are causing a risk that is independent to conventional ASCVD risk.
8. Inadequate data on non-rheumatoid arthritis autoimmune diseases with increased ASCVD risk.
9. No evidence on the relationship between anti-rheumatic drugs and ASCVD risk.
10. Inconclusive evidence on the efficacy of early preventative measures to reduce chemotherapy induced Type I cardiotoxicity.
11. Inadequate information of relative roles of life style interventions, exercise and calory restriction in the management of overweight and obesity.
12. Inadequate information on management and treatment goals of white coat hypertension.
13. No randomized trial evidence to decide on target levels to lower triglycerides.

References

1. Dhawan J, Bray CL, Warburton R, et al. Insulin resistance, high prevalence of diabetes, and cardiovascular risk in immigrant Asians. Genetic or environmental effect? *Br Heart J* 1994;72(5):413-421.
2. Anand SS, Yusuf S, Vuksan V, et al. Differences in risk factors, atherosclerosis, and cardiovascular disease between ethnic groups in Canada: the Study of Health Assessment and Risk in Ethnic groups (SHARE) *Lancet* 2000;356(9226):279-284.
3. Annual Health Bulletin – 2017. Ministry of Health, Sri Lanka.
4. Lloyd-Jones D, Adams R, Carnethon M, et al. Heart disease and stroke statistics—2009 update: a report from the American Heart Association Statistics Committee and Stroke Statistics Subcommittee. *Circulation*. 2009; 119:e21–e181.
5. Blood Pressure Lowering Treatment Trialists' Collaboration, Sundstrom J, Arima H, Woodward M, et al. Blood pressure-lowering treatment based on cardiovascular risk: a meta-analysis of individual patient data. *Lancet* 2014;384:591–598.
6. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 3. Effects in patients at different levels of cardiovascular risk—overview and meta-analyses of randomized trials. *J Hypertens* 2014;32:2305–2314.
7. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 2. Effects at different baseline and achieved blood pressure levels—overview and meta-analyses of randomized trials. *J Hypertens* 2014;32:2296–2304.
8. Goff DC Jr, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129:S49–S73.
9. Ference BA, Graham I, Tokgozoglul L, et al. Impact of lipids on cardiovascular health: JACC Health Promotion Series. *J Am Coll Cardiol*. 2018;72:1141–1156.
10. Anderson KM, Wilson PW, Odell PM, et al. An updated coronary risk profile. A statement for health professionals. *Circulation*. 1991;83:356–362.
11. D'Agostino RB, Grundy S, Sullivan LM, et al. Validation of the Framingham coronary heart disease prediction scores: results of a multiple ethnic groups investigation. *JAMA*. 2001; 286:180–187.
12. WHO Cardiovascular risk charts 2019. Available at: <https://www.who.int/news/item/02-09-2019-who-updates-cardiovascular-risk-charts>. Accessed June 15, 2021.
13. American College of Cardiology, American Heart Association. ASCVD Risk Estimator. Available at: https://tools.acc.org/ldl/ascvd_risk_estimator/index.html#!/calculate/estimator. Accessed October 31, 2020.

14. Conroy RM, Pyorala K, Fitzgerald AP, et al. Estimation of ten-year risk of fatal cardiovascular disease in Europe: the SCORE project. *Eur H J* 2003;24:987–1003.
15. Cappuccio FP, Oakeshott P, Strazzullo P, Kerry SM. Application of Framingham risk estimates to ethnic minorities in United Kingdom and implications for primary prevention of heart disease in general practice: cross sectional population-based study. *BMJ* 2002;325:1271.
16. Greenland P, Smith SC Jr, Grundy SM. Improving Coronary Heart Disease Risk Assessment in Asymptomatic People: Role of Traditional Risk Factors and Noninvasive Cardiovascular Tests. *Circulation*. 2001;104:1863–1867.
17. Bachmann JM, Willis BL, Ayers CR, et al. Association between family history and coronary heart disease death across long-term follow-up in men: the Cooper Center Longitudinal Study. *Circulation* 2012;125:3092–3098.
18. Banerjee A. A review of family history of cardiovascular disease: risk factor and research tool. *Int J Clin Pract* 2012;66:536–543.
19. World Health Organization. Familial hypercholesterolemia—report of a second WHO consultation. Geneva: World Health Organization, 1999.
20. Scientific Steering Committee on behalf of the Simon Broome Register Group. Risk of fatal coronary heart disease in familial hypercholesterolaemia. *BMJ* 1991;303:893–896.
21. Williams RR, Hunt SC, Schumacher MC, et al. Diagnosing heterozygous familial hypercholesterolemia using new practical criteria validated by molecular genetics. *Am J Cardiol* 1993;72:171–176.
22. Albert MA, Glynn RJ, Buring J, Ridker PM. Impact of traditional and novel risk factors on the relationship between socioeconomic status and incident cardiovascular events. *Circulation* 2006;114:2619–2626.
23. Alter DA, Franklin B, Ko DT, et al. Socioeconomic status, functional recovery, and long-term mortality among patients surviving acute myocardial infarction. *PLoS One* 2014;8:e65130.
24. Kivimaki M, Nyberg ST, Batty GD, et al. Job strain as a risk factor for coronary heart disease: a collaborative meta-analysis of individual participant data. *Lancet* 2012;380:1491–1497.
25. Eaker ED, Sullivan LM, Kelly-Hayes M, et al. Marital status, marital strain, and risk of coronary heart disease or total mortality: the Framingham Offspring Study. *Psychosom Med* 2007;69:509–513.
26. Spindler H, Pedersen SS. Posttraumatic stress disorder in the wake of heart disease: prevalence, risk factors, and future research directions. *Psychosom Med* 2005; 67:715–723.
26. Pogossova N, Saner H, Pedersen SS, et al. Psychosocial aspects in cardiac rehabilitation: from theory to practice. A position paper from the Cardiac Rehabilitation Section of the European Association of Cardiovascular Prevention and Rehabilitation of the European Society of Cardiology. *Eur J Prev Cardiol* 2015;22:1290–1306.
27. Fan Z, Wu Y, Shen J, Ji T, Zhan R. Schizophrenia and the risk of cardiovascular diseases: a meta-analysis of thirteen cohort studies. *J Psychiatr Res* 2013;47:1549–1556.

28. Chida Y, Steptoe A. The association of anger and hostility with future coronary heart disease: a meta-analytic review of prospective evidence. *J Am Coll Cardiol* 2009;53:936–946.
29. Barth J, Schneider S, von Kanel R. Lack of social support in the etiology and the prognosis of coronary heart disease: a systematic review and meta-analysis. *Psychosom Med* 2010;72:229–238.
30. Albus C, Jordan J, Herrmann-Lingen C. Screening for psychosocial risk factors in patients with coronary heart disease-recommendations for clinical practice. *Eur J Cardiovasc Prev Rehabil* 2004;11:819–28.
31. Wilson BJ, Watson MS, Prescott GJ, et al. Hypertensive diseases of pregnancy and risk of hypertension and stroke in later life: results from cohort study. *BMJ* 2003;326:845.
32. Lykke JA, Langhoff-Roos J, Sibai BM, et al. Hypertensive pregnancy disorders and subsequent cardiovascular morbidity and type 2 diabetes mellitus in the mother. *Hypertension* 2009;53:944–951.
33. Lykke JA, Paidas MJ, Damm P, et al. Preterm delivery and risk of subsequent cardiovascular morbidity and type-II diabetes in the mother. *BJOG* 2010;117:274–281.
34. Bellamy L, Casas JP, Hingorani AD, Williams D. Type 2 diabetes mellitus after gestational diabetes: a systematic review and meta-analysis. *Lancet* 2009;373:1773–1779.
35. Muka T, Oliver-Williams C, Kunutsor S, et al. Association of age at onset of menopause and time since onset of menopause with cardiovascular outcomes, intermediate vascular traits, and all-cause mortality: a systematic review and meta-analysis. *JAMA Cardiol* 2016;1:767–776.
36. Morgan CL, Jenkins-Jones S, Currie CJ, Rees DA. Evaluation of adverse outcome in young women with polycystic ovary syndrome versus matched, reference controls: a retrospective, observational study. *J Clin Endocrinol Metab* 2012;97:3251–3260.
37. Schiffrin EL, Lipman ML, Mann JF. Chronic kidney disease: effects on the cardiovascular system. *Circulation* 2007;116:85–97.
38. Perkovic V, Verdon C, Ninomiya T, et al. The relationship between proteinuria and coronary risk: a systematic review and meta-analysis. *PLoS Med*. 2008;5:e207.
39. Lambers Heerspink HJ, Brantsma AH, De Zeeuw D, et al. Albuminuria assessed from first-morning-void urine samples versus 24-hour urine collections as a predictor of cardiovascular morbidity and mortality. *Am J Epidemiol*. 2008;168:897–905.
40. Levey AS, Eckardt KU, Tsukamoto Y, et al. Definition and classification of chronic kidney disease: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int*. 2005;67:2089–2100.
41. Astor BC, Hallan SI, Miller ER 3rd, et al. Glomerular filtration rate, albuminuria, and risk of cardiovascular and all-cause mortality in the US population. *Am J Epidemiol*. 2008;167:1226–1234.
42. Cao JJ, Biggs ML, Barzilay J, et al. Cardiovascular and mortality risk prediction and stratification using urinary albumin excretion in older adults ages 68–102: the Cardiovascular Health Study. *Atherosclerosis*. 2008;197:806–813.

43. Barzilay JI, Forsberg C, Heckbert SR, et al. The association of markers of inflammation with weight change in older adults: the Cardiovascular Health Study. *Int J Obes (Lond)*. 2006;30:1362–1367.
44. Pradhan AD, Manson JE, Rifai N, et al. C-reactive protein, interleukin 6, and risk of developing type 2 diabetes mellitus. *JAMA*. 2001;286:327–334.
45. Sesso HD, Buring JE, Rifai N, et al. C-reactive protein and the risk of developing hypertension. *JAMA*. 2003;290:2945–2951.
46. Pradhan AD, Rifai N, Ridker PM. Soluble intercellular adhesion molecule-1, soluble vascular adhesion molecule-1, and the development of symptomatic peripheral arterial disease in men. *Circulation*. 2002;106:820–825.
47. Gottdiener JS, Arnold AM, Aurigemma GP, et al. Predictors of congestive heart failure in the elderly: the Cardiovascular Health Study. *J Am Coll Cardiol*. 2000;35:1628–1637.
48. Aviles RJ, Martin DO, Apperson-Hansen C, et al. Inflammation as a risk factor for atrial fibrillation. *Circulation*. 2003;108:3006–3010.
49. Albert CM, Ma J, Rifai N, et al. Prospective study of C-reactive protein, homocysteine, and plasma lipid levels as predictors of sudden cardiac death. *Circulation*. 2002;105:2595–2599.
50. Harris TB, Ferrucci L, Tracy RP, et al. Associations of elevated interleukin-6 and C-reactive protein levels with mortality in the elderly. *Am J Med*. 1999;106:506–512.
51. Tinana A, Mintz GS, Weissman NJ. Volumetric intravascular ultrasound quantification of the amount of atherosclerosis and calcium in non-stenotic arterial segments. *Am J Cardiol* 2002;89:757–760.
52. Nasir K, Budoff MJ, Post WS, et al. Electron beam CT versus helical CT scans for assessing coronary calcification: current utility and future directions. *Am Heart J*. 2003;146:969–977.
53. Silber S. Comparison of spiral and electron beam tomography in the evaluation of coronary calcification in asymptomatic persons. *Int J Cardiol* 2002;82:297–298; author reply 299.
54. Budoff MJ, Achenbach S, Blumenthal RS, et al. Assessment of coronary artery disease by cardiac computed tomography: a scientific statement from the American Heart Association Committee on Cardiovascular Imaging and Intervention, Council on Cardiovascular Radiology and Intervention, and Committee on Cardiac Imaging, Council on Clinical Cardiology. *Circulation*. 2006;114:1761–1791.
55. Morin RL, Gerber TC, McCollough CH. Radiation dose in computed tomography of the heart. *Circulation*. 2003;107:917–922.
56. Kim KP, Einstein AJ, Berrington de GA. Coronary artery calcification screening: estimated radiation dose and cancer risk. *Arch Intern Med*. 2009;169:1188–1194.
57. Greenland P, LaBree L, Azen SP, et al. Coronary artery calcium score combined with Framingham score for risk prediction in asymptomatic individuals. *JAMA*. 2004;291:210–215.

58. Budoff MJ, Nasir K, McClelland RL, et al. Coronary calcium predicts events better with absolute calcium scores than age-sex-race/ethnicity percentiles: MESA (Multi-Ethnic Study of Atherosclerosis). *J Am Coll Cardiol*. 2009;53:345–352.
59. Hippisley-Cox J, Coupland C, Vinogradova Y, et al. Predicting cardiovascular risk in England and Wales: prospective derivation and validation of QRISK2. *BMJ* 2008;336:1475–1482.
60. Ogdie A, Yu Y, Haynes K, Love TJ, et al. Risk of major cardiovascular events in patients with psoriatic arthritis, psoriasis and rheumatoid arthritis: a population-based cohort study. *Ann Rheum Dis* 2015;74:326–332.
61. Sattar N, McCarey DW, Capell H, McInnes IB. Explaining how “high-grade” systemic inflammation accelerates vascular risk in rheumatoid arthritis. *Circulation* 2003;108:2957–2963.
62. Peters MJ, Symmons DP, McCarey D, et al. EULAR evidence-based recommendations for cardiovascular risk management in patients with rheumatoid arthritis and other forms of inflammatory arthritis. *Ann Rheum Dis* 2010;69:325–331.
63. Montorsi P, Ravagnani PM, Galli S, et al. Association between erectile dysfunction and coronary artery disease: matching the right target with the right test in the right patient. *Eur Urol* 2006;50:721–731.
64. Vlachopoulos CV, Terentes-Printzios DG, Ioakeimidis NK, et al. Prediction of cardiovascular events and all-cause mortality with erectile dysfunction: a systematic review and meta-analysis of cohort studies. *Circ Cardiovasc Qual Outcomes* 2013;6:99–109.
65. Hiatt WR. Medical treatment of peripheral arterial disease and claudication. *N Engl J Med* 2001;344:1608–1621.
66. Fowkes FG, Price JF, Stewart MC, et al. Aspirin for prevention of cardiovascular events in a general population screened for a low ankle brachial index: a randomized controlled trial. *JAMA* 2010;303:841–848.
67. Fowkes FG, Murray GD, Butcher I, et al. Ankle brachial index combined with Framingham Risk Score to predict cardiovascular events mortality: a meta-analysis. *JAMA*. 2008;300:197–208.
68. Hooi JD, Kester AD, Stoffers HE, et al. Asymptomatic peripheral arterial occlusive disease predicted cardiovascular morbidity and mortality in a 7-year follow-up study. *J Clin Epidemiol*. 2004;57:294 –300.
69. O’Leary DH, Polak JF, Kronmal RA, et al. Carotid-artery intima and media thickness as a risk factor for myocardial infarction and stroke in older adults. Cardiovascular Health Study Collaborative Research Group. *N Engl J Med* 1999;340:14–22.
70. Den Ruijter HM, Peters SA, Anderson TJ, et al. Common carotid intima-media thickness measurements in cardiovascular risk prediction: a meta-analysis. *JAMA* 2012;308:796–803.
71. Chambless LE, Heiss G, Folsom AR, Rosamond W, et al. Association of coronary heart disease incidence with carotid arterial wall thickness and major risk factors: the Atherosclerosis Risk in Communities (ARIC) Study, 1987–1993. *Am J Epidemiol* 1997;146:483–494.

72. Ashley EA, Raxwal V, Froelicher V. An evidence-based review of the resting electrocardiogram as a screening technique for heart disease. *Prog Cardiovasc Dis.* 2001;44:55– 67.
73. Denes P, Larson JC, Lloyd-Jones DM, et al. Major and minor ECG abnormalities in asymptomatic women and risk of cardiovascular events and mortality. *JAMA.* 2007;297:978–985.
74. Okin PM, Devereux RB, Nieminen MS, et al. Electrocardiographic strain pattern and prediction of cardiovascular morbidity and mortality in hypertensive patients. *Hypertension.* 2004;44:48 –54.
75. Gorodeski EZ, Ishwaran H, Blackstone EH, Lauer MS. Quantitative electrocardiographic measures and long-term mortality in exercise test patients with clinically normal resting electrocardiograms. *Am Heart J.* 2009;158:61–70.
76. American Diabetes Association. Standards of medical care in diabetes—2010. *Diabetes Care.* 2010;33 Suppl 1:S11–S61.
77. Adams RJ, Appleton SL, Hill CL, et al. Independent association of HbA(1c) and incident cardiovascular disease in people without diabetes. *Obesity.* 2009;17:559–563.
78. Khaw KT, Wareham N, Bingham S, et al. Association of hemoglobin A1c with cardiovascular disease and mortality in adults: the European prospective investigation into cancer in Norfolk. *Ann Intern Med.* 2004;141:413–420.
79. Selvin E, Steffes MW, Zhu H, et al. Glycated hemoglobin, diabetes, and cardiovascular risk in nondiabetic adults. *N Engl J Med.* 2010;362:800–811.
80. Lauer M, Froelicher ES, Williams M, et al. Exercise testing in asymptomatic adults: a statement for professionals from the American Heart Association Council on Clinical Cardiology, Subcommittee on Exercise, Cardiac Rehabilitation, and Prevention. *Circulation.* 2005;112:771–776.
81. Nishime EO, Cole CR, Blackstone EH, et al. Heart rate recovery and treadmill exercise score as predictors of mortality in patients referred for exercise ECG. *JAMA.* 2000;284:1392–1398.
82. Mark DB, Shaw L, Harrell FE Jr, et al. Prognostic value of a treadmill exercise score in outpatients with suspected coronary artery disease. *N Engl J Med.* 1991;325:849–853.
83. Gulati M, Pandey DK, Arnsdorf MF, et al. Exercise capacity and the risk of death in women: the St James Women Take Heart Project. *Circulation.* 2003;108:1554–1559.
84. Adabag AS, Grandits GA, Prineas RJ, et al. Relation of heart rate parameters during exercise test to sudden death and all-cause mortality in asymptomatic men. *Am J Cardiol.* 2008;101:1437–1443.
85. Wei M, Kampert JB, Barlow CE, et al. Relationship between low cardiorespiratory fitness and mortality in normal-weight, overweight, and obese men. *JAMA.* 1999;282:1547–1553.
86. Levy D, Garrison RJ, Savage DD, et al. Prognostic implications of echocardiographically determined left ventricular mass in the Framingham Heart Study. *N Engl J Med.* 1990;322:1561–1566.
87. Nunez E, Arnett DK, Benjamin EJ, et al. Comparison of the prognostic value of left ventricular hypertrophy in African-American men versus women. *Am J Cardiol.* 2004;94:1383–1390.

88. Okin PM, Roman MJ, Lee ET, et al. Combined echocardiographic left ventricular hypertrophy and electrocardiographic ST depression improve prediction of mortality in American Indians: the Strong Heart Study. *Hypertension*. 2004;43:769–774.
89. Verdecchia P, Carini G, Circo A, et al. Left ventricular mass and cardiovascular morbidity in essential hypertension: the MAVI study. *J Am Coll Cardiol*. 2001;38:1829–1835.
90. Rodriguez CJ, Lin F, Sacco RL, et al. Prognostic implications of left ventricular mass among Hispanics: the Northern Manhattan Study. *Hypertension*. 2006;48:87–92.
91. Berman DS, Rozanski A, Rana JS, et al. Screening for coronary artery disease in diabetic patients: a commentary. *J Nucl Cardiol*. 2009;16:851–4.
92. Einstein AJ, Moser KW, Thompson RC, et al. Radiation dose to patients from cardiac diagnostic imaging. *Circulation*. 2007;116:1290–1305.
93. Berman DS, Wong ND, Gransar H, et al. Relationship between stress-induced myocardial ischemia and atherosclerosis measured by coronary calcium tomography. *J Am Coll Cardiol*. 2004;44:923–930.
94. Anand DV, Lim E, Hopkins D, et al. Risk stratification in uncomplicated type 2 diabetes: prospective evaluation of the combined use of coronary artery calcium imaging and selective myocardial perfusion scintigraphy. *Eur Heart J*. 2006;27:713–721.
95. Blumenthal RS, Becker DM, Yanek LR, et al. Comparison of coronary calcium and stress myocardial perfusion imaging in apparently healthy siblings of individuals with premature coronary artery disease. *Am J Cardiol*. 2006;97:328–333.
96. Gonzaga C, Bertolami A, Bertolami M, et al. Obstructive sleep apnea, hypertension and cardiovascular diseases. *J Hum Hypertens* 2015;29:705–712.
97. Offit K. Genomic profiles for disease risk: predictive or premature? *JAMA*. 2008;299:1353–1355.
98. Baccarelli A, Wright R, Bollati V, et al. Ischemic heart disease and stroke in relation to blood DNA methylation. *Epidemiology* 2010;21:819–828.
99. DiAngelantonio E, Chowdhury R, Sarwar N, et al. B-type natriuretic peptides and cardiovascular risk: systematic review and meta-analysis of 40 prospective studies. *Circulation*. 2009;120:2177–2187.
100. Zalewski A, Macphee C. Role of lipoprotein-associated phospholipase A2 in atherosclerosis: biology, epidemiology, and possible therapeutic target. *Arterioscler Thromb Vasc Biol*. 2005;25:923–931.
101. Garza CA, Montori VM, McConnell JP, et al. Association between lipoprotein-associated phospholipase A2 and cardiovascular disease: a systematic review. *Mayo Clin Proc*. 2007;82:159–165.
102. Vlachopoulos C, Aznaouridis K, Stefanadis C. Prediction of cardiovascular events and all-cause mortality with arterial stiffness: a systematic review and meta-analysis. *J Am Coll Cardiol* 2010;55:1318–1327.
103. Kullo IJ, Malik AR. Arterial ultrasonography and tonometry as adjuncts to cardiovascular risk stratification. *J Am Coll Cardiol*. 2007;49:1413–1426.

104. Bluemke DA, Achenbach S, Budoff M, et al. Noninvasive coronary artery imaging: magnetic resonance angiography and multidetector computed tomography angiography: a scientific statement from the American Heart Association Committee on Cardiovascular Imaging and Intervention of the Council on Cardiovascular Radiology and Intervention and the Councils on Clinical Cardiology and Cardiovascular Disease in the Young. *Circulation*. 2008;118:586–606.
105. Choi EK, Choi SI, Rivera JJ, et al. Coronary computed tomography angiography as a screening tool for the detection of occult coronary artery disease in asymptomatic individuals. *J Am Coll Cardiol*. 2008;52:357–365.
106. Fleisher LA, Beckman JA, Brown KA, et al. ACC/AHA 2007 guidelines on perioperative cardiovascular evaluation and care for noncardiac surgery: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Writing Committee to Revise the 2002 Guidelines on Perioperative Cardiovascular Evaluation for Noncardiac Surgery). *J Am Coll Cardiol*. 2007;50:e159–e241.
107. Choudhury RP, Fuster V, Badimon JJ, et al. MRI and characterization of atherosclerotic plaque: emerging applications and molecular imaging. *Arterioscler Thromb Vasc Biol*. 2002;22:1065–1074.
108. Fayad ZA, Fuster V. Clinical imaging of the high-risk or vulnerable atherosclerotic plaque. *Circ Res*. 2001;89:305–316.
109. Jaffer FA, Tung CH, Gerszten RE, Weissleder R. In vivo imaging of thrombin activity in experimental thrombi with thrombin-sensitive near-infrared molecular probe. *Arterioscler Thromb Vasc Biol*. 2002;22:1929–1935.
110. El Harchaoui K, van der Steeg WA, Stroes ES, et al. Value of low-density lipoprotein particle number and size as predictors of coronary artery disease in apparently healthy men and women: the EPIC-Norfolk Prospective Population Study. *J Am Coll Cardiol*. 2007;49:547–553.
111. Ip S, Lichtenstein AH, Chung M, et al. Systematic review: association of low-density lipoprotein subfractions with cardiovascular outcomes. *Ann Intern Med*. 2009;150:474–484.
112. Ingelsson E, Schaefer EJ, Contois JH, et al. Clinical utility of different lipid measures for prediction of coronary heart disease in men and women. *JAMA*. 2007;298:776–785.
113. Erqou S, Kaptoge S, Perry PL, et al. Lipoprotein(a) concentration and the risk of coronary heart disease, stroke, and nonvascular mortality. *JAMA*. 2009;302:412–423.
114. Dati F, Tate JR, Marcovina SM, Steinmetz A. First WHO/IFCC International Reference Reagent for Lipoprotein(a) for Immunoassay—Lp(a) SRM 2B. *Clin Chem Lab Med*. 2004;42:670–676.
115. European Heart Network. Diet, Physical Activity and Cardiovascular Disease Prevention in Europe. Brussels, Belgium: European Heart Network, 2011.
116. Astrup A, Dyerberg J, Elwood P, et al. The role of reducing intakes of saturated fat in the prevention of cardiovascular disease: where does the evidence stand in 2010? *Am J Clin Nutr* 2011;93:684–688.
117. Rizos EC, Ntzani EE, Bika E, et al. Association between omega-3 fatty acid supplementation and risk of major cardiovascular disease events: a systematic review and meta-analysis. *JAMA* 2012;308:1024–33.

118. Mozaffarian D, Katan MB, Ascherio A, et al. Trans fatty acids and cardiovascular disease. *N Engl J Med* 2006;354:1601–1613.
119. Sacks FM, Svetkey LP, Vollmer WM, et al. Effects on blood pressure of reduced dietary sodium the Dietary Approaches to Stop Hypertension (DASH) diet. DASH-Sodium Collaborative Research Group. *N Engl J Med* 2001;344:3–10.
120. Aburto NJ, Hanson S, Gutierrez H, et al. Effect of increased potassium intake on cardiovascular risk factors and disease: systematic review and meta-analyses. *BMJ* 2013;346:f1378.
121. Chowdhury R, Stevens S, Gorman D, et al. Association between fish consumption, long chain omega 3 fatty acids, and risk of cerebrovascular disease: systematic review and meta-analysis. *BMJ* 2012;345:e6698.
122. Threapleton DE, Greenwood DC, Evans CE, et al. Dietary fibre intake and risk of cardiovascular disease: systematic review and meta-analysis. *BMJ* 2013;347:f6879.
123. Luo C, Zhang Y, Ding Y, et al. Nut consumption and risk of type 2 diabetes, cardiovascular disease, and all-cause mortality: a systematic review and meta-analysis. *Am J Clin Nutr* 2014;100:256–269.
124. Wang X, Ouyang Y, Liu J, et al. Fruit and vegetable consumption and mortality from all causes, cardiovascular disease, and cancer: systematic review and dose-response meta-analysis of prospective cohort studies. *BMJ* 2014;349:g4490.
125. Zheng J, Huang T, Yu Y, et al. Fish consumption and CHD mortality: an updated meta-analysis of seventeen cohort studies. *Public Health Nutr* 2012;15:725–737.
126. Chowdhury R, Kunutsor S, Vitezova A, et al. Vitamin D and risk of cause specific death: systematic review and meta-analysis of observational cohort and randomized intervention studies. *BMJ* 2014;348:g1903.
127. Rizos EC, Ntzani EE, Bika E, et al. Association between omega-3 fatty acid supplementation and risk of major cardiovascular disease events: a systematic review and meta-analysis. *JAMA* 2012;308:1024–1033.
128. Burton, R., & Sheron, N. (2018). No level of alcohol consumption improves health. *The Lancet*, 392(10152), 987–988. [https://doi.org/10.1016/S0140-6736\(18\)31571-X](https://doi.org/10.1016/S0140-6736(18)31571-X)
129. World Health Organization. Guideline: Sugars Intake for Adults and Children. Geneva: World Health Organization, 2015.
130. Physical Activity Guidelines Advisory Committee. Physical Activity Guidelines Advisory Committee Report, 2008. Washington, DC: U.S. Department of Health and Human Services, 2008.
131. Talbot LA, Morrell CH, Fleg JL, et al. Changes in leisure time physical activity and risk of all-cause mortality in men and women: the Baltimore Longitudinal Study of Aging. *Prev Med* 2007;45:169–176.
132. Sattelmair J, Pertman J, Ding EL, et al. Dose response between physical activity risk of coronary heart disease: a meta-analysis. *Circulation* 2011;124:789–795.
133. Moore SC, Patel AV, Matthews CE, et al. Leisure time physical activity of moderate to vigorous intensity and mortality: a large pooled cohort analysis. *PLoS Med* 2012;9:e1001335.

134. Lee DC, Pate RR, Lavie CJ, et al. Leisure-time running reduces all-cause and cardiovascular mortality risk. *J Am Coll Cardiol* 2014;64:472–481.
135. Lollgen H, Bockenhoff A, Knapp G. Physical activity and all-cause mortality: an updated meta-analysis with different intensity categories. *Int J Sports Med* 2009;30:213–224.
136. Kyu HH, Bachman VF, Alexander LT, et al. Physical activity and risk of breast cancer, colon cancer, diabetes, ischemic heart disease, and ischemic stroke events: systematic review and dose-response meta-analysis for the Global Burden of Disease Study 2013. *BMJ*. 2016;354:i3857.
137. Sattelmair J, Pertman J, Ding EL, et al. Dose response between physical activity and risk of coronary heart disease: a meta-analysis. *Circulation*. 2011;124:789–95.
138. Borjesson M, Urhausen A, Kouidi E, et al. Cardiovascular evaluation of middle-aged/senior individuals engaged in leisure-time sport activities: position stand from the sections of exercise physiology and sports cardiology of the European Association of Cardiovascular Prevention and Rehabilitation. *Eur J Cardiovasc Prev Rehabil* 2011;18:446–458.
139. Physical Activity Guidelines Advisory Committee. Physical Activity Guidelines Advisory Committee Report, 2008. Washington, DC: U.S. Department of Health and Human Services, 2008.
140. Centers for Disease Control and Prevention. How Tobacco Smoke Causes Disease: The Biology and Behavioural Basis for Smoking-attributable Disease 2010. A Report of the Surgeon General. <http://www.surgeongeneral.gov/library/tobaccosmoke/index.html>.
141. Doll R, Peto R, Boreham J, et al. Mortality in relation to smoking: 50 years' observations on male British doctors. *BMJ* 2004;328:1519.
142. Prescott E, Scharling H, Osler M, et al. Importance of light smoking and inhalation habits on risk of myocardial infarction and all-cause mortality. A 22 year follow up of 12 149 men and women in the Copenhagen City Heart Study. *J Epidemiol Community Health* 2002;56:702–706.
143. Law MR, Morris JK, Wald NJ. Environmental tobacco smoke exposure and ischaemic heart disease: an evaluation of the evidence. *BMJ* 1997;315:973–980.
144. Joseph AM, Norman SM, Ferry LH, et al. The safety of transdermal nicotine as an aid to smoking cessation in patients with cardiac disease. *N Engl J Med* 1996;335:1792–1798.
145. Cahill K, Stead LF, Lancaster T. Nicotine receptor partial agonists for smoking cessation. *Cochrane Database Syst Rev* 2012;4:CD006103.
146. Rajasuriya, M. (2017). A review of interventions for smoking cessation. *Sri Lanka Journal of Psychiatry*, 8(1), 10. <https://doi.org/10.4038/slipsyc.v8i1.8133>
147. Bullen C, Howe C, Laugesen M, et al. Electronic cigarettes for smoking cessation: a randomized controlled trial. *Lancet* 2013;382:1629–1637.
148. Henningfield JE. Nicotine medications for smoking cessation. *N Engl J Med* 1995;333:1196–1203.

149. Passenger C, Dossing M. A systematic review of health effects of electronic cigarettes. *Prev Med* 2014;69:248–260.
150. LeBlanc EL, Patnode CD, Webber EM, et al. Draft evidence review for weight loss to prevent obesity-related morbidity and mortality in adults: behavioral interventions. Kaiser Permanente Research Affiliates Evidence-based Practice Center, Kaiser Permanente Center for Health Research: Portland, OR; 2018. AHRQ Publication No. 18-05239-EF-1. Available at: <https://www.uspreventiveservicestaskforce.org/Page/Document/draft-evidence-review/obesity-in-adultsinterventions1>. Accessed on: November 10, 2020.
151. Canoy D, Cairns BJ, Balkwill A, et al. Coronary heart disease incidence in women by waist circumference within categories of body mass index. *Eur J Prev Cardiol*. 2013;20:759–762.
152. Obesity in Asia Collaboration. Waist Circumference Thresholds Provide an Accurate and Widely Applicable Method for the Discrimination of Diabetes. *Diabetes Care* 2007 Dec; 30(12): 3116-3118.
153. Ma C, Avenell A, Bolland M, et al. Effects of weight loss interventions for adults who are obese on mortality, cardiovascular disease, and cancer: systematic review and meta-analysis. *BMJ*. 2017;359:j4849.
154. Kwok CS, Pradhan A, Khan MA, et al. Bariatric surgery and its impact on cardiovascular disease and mortality: a systematic review and meta-analysis. *Int J Cardiol* 2014;173:20–28.
155. Emerging Risk Factors Collaboration, Sarwar N, Gao P, Seshasai SR, Gobin R, et al. Diabetes mellitus, fasting blood glucose concentration, and risk of vascular disease: a collaborative meta-analysis of 102 prospective studies. *Lancet* 2010;375:2215–2222.
156. Ryden L, Grant PJ, Anker SD, et al. ESC guidelines on diabetes, pre-diabetes, and cardiovascular diseases developed in collaboration with the EASD: the Task Force on diabetes, pre-diabetes, and cardiovascular diseases of the European Society of Cardiology (ESC) and developed in collaboration with the European Association for the Study of Diabetes (EASD). *Eur Heart J* 2013;34:3035–3087.
157. UK Prospective Diabetes Study (UKPDS) Group. Intensive blood-glucose control with sulphonylureas or insulin compared with conventional treatment and risk of complications in patients with type 2 diabetes (UKPDS 33). *Lancet* 1998;352:837–853.
158. The ADVANCE Collaborative Group, Patel A, MacMahon S, Chalmers J, et al. Intensive blood glucose control and vascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2008;358:2560–2572.
159. Pan A, Wang Y, Talaei M, et al. Relation of active, passive, and quitting smoking with incident type 2 diabetes: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol* 2015;3:958–967.
160. Wannamethee SG, Shaper AG, Whincup PH, et al. Impact of diabetes on cardiovascular disease risk and all-cause mortality in older men: influence of age at onset, diabetes duration, and established and novel risk factors. *Arch Intern Med* 2011;171:404–410.
161. Preiss D, Sattar N, McMurray JJ. A systematic review of event rates in clinical trials in diabetes mellitus: the importance of quantifying baseline cardiovascular disease history and proteinuria and implications for clinical trial design. *Am Heart J* 2011;161:p210–219.e1.

162. Tonelli M, Muntner P, Lloyd A, et al. Risk of coronary events in people with chronic kidney disease compared with those with diabetes: a population-level cohort study. *Lancet* 2012;380:807–814.
163. Dormandy JA, Charbonnel B, Eckland DJ, et al. Secondary prevention of macrovascular events in patients with type 2 diabetes in the PROactive Study (PROspective pioglitAzone Clinical Trial In macroVascular Events): a randomized controlled trial. *Lancet* 2005;366:1279–1289.
164. Kelly TN, Bazzano LA, Fonseca VA, et al. Systematic review: glucose control and cardiovascular disease in type 2 diabetes. *Ann Intern Med* 2009;151:394–403.
165. UK Prospective Diabetes Study (UKPDS) Group. Effect of intensive blood glucose control with metformin on complications in overweight patients with type 2 diabetes (UKPDS 34). *Lancet* 1998;352:854–865.
166. Effect of intensive blood-glucose control with metformin on complications in overweight patients with type 2 diabetes (UKPDS 34). UK Prospective Diabetes Study (UKPDS) Group. *Lancet*. 1998;352:854–865.
167. Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med* 2015;373:2117–2128.
168. Neal B, Perkovic V, Mahaffey KW, et al. Canagliflozin and cardiovascular and renal events in type 2 diabetes. *N Engl J Med*. 2017;377:644–657.
169. Wiviott SD, Raz I, Bonaca MP, et al. Dapagliflozin and cardiovascular outcomes in type 2 diabetes. *N Engl J Med*. 2019;380:347–357.
170. Marso SP, Daniels GH, Brown-Frandsen K, et al. Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med*. 2016;375:311–322.
171. Hernandez AF, Green JB, Janmohamed S, et al. Albiglutide and cardiovascular outcomes in patients with type 2 diabetes and cardiovascular disease (Harmony Outcomes): a double-blind, randomized placebo-controlled trial. *Lancet*. 2018;392:1519–1529.
172. Marso SP, Bain SC, Consoli A, et al. Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med*. 2016;375:1834–1844.
173. Collins R, Armitage J, Parish S, et al. MRC/BHF Heart Protection Study of cholesterol-lowering with simvastatin in 5963 people with diabetes: a randomized placebo-controlled trial. *Lancet* 2003;361:2005–2016.
174. Colhoun HM, Betteridge DJ, Durrington PN, et al. Primary prevention of cardiovascular disease with atorvastatin in type 2 diabetes in the Collaborative Atorvastatin Diabetes Study (CARDS): multicenter randomized placebo-controlled trial. *Lancet* 2004;364:685–696.
175. Shepherd J, Barter P, Carmena R, et al. Effect of lowering LDL cholesterol substantially below currently recommended levels in patients with coronary heart disease and diabetes: the Treating to New Targets (TNT) study. *Diabetes Care* 2006;29:1220–1226.
176. Cannon CP, Blazing MA, Giugliano RP, et al. Ezetimibe added to statin therapy after acute coronary syndromes. *N Engl J Med* 2015;372:2387–2397.

177. Emdin CA, Rahimi K, Neal B, et al. Blood pressure lowering in type 2 diabetes: a systematic review and meta-analysis. *JAMA* 2015;313:603–615.
178. Lind M, Svensson AM, Kosiborod M, et al. Glycemic control and excess mortality in type 1 diabetes. *N Engl J Med* 2014;371:1972–1982.
179. Lewington S, Clarke R, Qizilbash N, et al. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet* 2002;360:1903–1913.
180. Fagard RH. Exercise therapy in hypertensive cardiovascular disease. *Prog Cardiovasc Dis* 2011;53:404–411.
181. Cook NR, Cutler JA, Obarzanek E, et al. Long-term effects of dietary sodium reduction on cardiovascular disease outcomes: observational follow-up of the trials of hypertension prevention (TOHP). *BMJ* 2007;334:885–888.
182. Estruch R, Ros E, Salas-Salvado J, et al. Primary prevention of cardiovascular disease with a Mediterranean diet. *N Engl J Med* 2013;368:1279–1290.
183. ESH/ESC Task Force for the Management of Arterial Hypertension. 2013 Practice guidelines for the management of arterial hypertension of the European Society of Hypertension (ESH) and the European Society of Cardiology (ESC): ESH/ESC Task Force for the Management of Arterial Hypertension. *J Hypertens* 2013;31:1925–1938.
184. Law MR, Morris JK, Wald NJ. Use of blood pressure lowering drugs in the prevention of cardiovascular disease: meta-analysis of 147 randomized trials in the context of expectations from prospective epidemiological studies. *BMJ* 2009;338:b1665.
185. Zanchetti A, Mancia G. Longing for clinical excellence: a critical outlook into the NICE recommendations on hypertension management—is nice always good? *J Hypertens* 2012;30:660–668.
186. Blood Pressure Lowering Treatment Trialists' Collaboration, Turnbull F, Neal B, Ninomiya T, et al. Effects of different regimens to lower blood pressure on major cardiovascular events in older and younger adults: meta-analysis of randomized trials. *BMJ* 2008;336:1121–1123.
187. Wald DS, Law M, Morris JK, et al. Combination therapy versus monotherapy in reducing blood pressure: meta-analysis on 11,000 participants from 42 trials. *Am J Med* 2009;122:290–300.
188. Mancia G, Fagard R, Narkiewicz K, et al. 2013 ESH/ESC practice guidelines for the management of arterial hypertension. *Blood Pressure* 2014;23:3–16.
189. Zanchetti A, Grassi G, Mancia G. When should antihypertensive drug treatment be initiated and to what levels should systolic blood pressure be lowered? A critical reappraisal. *J Hypertens* 2009;27:923–934.
190. Ettehad D, Emdin CA, Kiran A, et al. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and metanalysis. *Lancet*. 2016;387:957–967.
191. Xie X, Atkins E, Lv J, et al. Effects of intensive blood pressure lowering on cardiovascular and renal outcomes: updated systematic review and meta-analysis. *Lancet*. 2016;387:435–443.

192. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 7. Effects of more vs. less intensive blood pressure lowering and different achieved blood pressure levels—updated overview and meta-analyses of randomized trials. *J Hypertens*. 2016;34:613–622.
193. Neaton JD, Blackburn H, Jacobs D, et al. Serum cholesterol level and mortality findings for men screened in the Multiple Risk Factor Intervention Trial. Multiple Risk Factor Intervention Trial Research Group. *Arch Intern Med* 1992;152:1490–1500.
194. Cholesterol Treatment Trialists' Collaboration, Mihaylova B, Emberson J, Blackwell L, et al. The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease: meta-analysis of individual data from 27 randomized trials. *Lancet* 2012;380:581–590.
195. Robinson JG, Wang S, Smith BJ, et al. Meta-analysis of the relationship between non-high-density lipoprotein cholesterol reduction and coronary heart disease risk. *J Am Coll Cardiol* 2009;53:316–322.
196. Chapman MJ, Ginsberg HN, Amarencu P, et al. Triglyceride-rich lipoproteins and high-density lipoprotein cholesterol in patients at high risk of cardiovascular disease: evidence and guidance for management. *Eur Heart J* 2011;32:1345–1361.
197. Sarwar N, Danesh J, Eiriksdottir G, et al. Triglycerides and the risk of coronary heart disease: 10,158 incident cases among 262,525 participants in 29 Western prospective studies. *Circulation* 2007;115:450–458.
198. Colhoun HM, Betteridge DJ, Durrington PN, et al. Primary prevention of cardiovascular disease with atorvastatin in type 2 diabetes in the Collaborative Atorvastatin Diabetes Study (CARDS): multicenter randomized placebo-controlled trial. *Lancet* 2004;364:685–696.
199. Collins R, Armitage J, Parish S, et al. MRC/BHF Heart Protection Study of cholesterol-lowering with simvastatin in 5963 people with diabetes: a randomized placebo-controlled trial. *Lancet* 2003;361:2005–2016.
200. Colhoun HM, Betteridge DJ, Durrington PN, et al. Primary prevention of cardiovascular disease with atorvastatin in type 2 diabetes in the Collaborative Atorvastatin Diabetes Study (CARDS): multicenter randomized placebo-controlled trial. *Lancet*. 2004;364:685–696.
201. Collins R, Armitage J, Parish S, et al. MRC/BHF Heart Protection Study of cholesterol-lowering with simvastatin in 5963 people with diabetes: a randomized placebo-controlled trial. *Lancet*. 2003;361:2005–2016.
202. Knopp RH, d'Emden M, Smilde JG, et al. Efficacy and safety of atorvastatin in the prevention of cardiovascular end points in subjects with type 2 diabetes: the Atorvastatin Study for Prevention of Coronary Heart Disease Endpoints in non-insulin dependent diabetes mellitus (ASPEN). *Diabetes Care*. 2006;29:1478–1485.
203. Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018 Nov, 73 (24) e285–e350.

204. Cholesterol Treatment Trialists' (CTT) Collaboration, Baigent C, Blackwell L, et al. Efficacy and safety of more intensive lowering of LDL cholesterol: a meta-analysis of data from 170,000 participants in 26 randomized trials. *Lancet*. 2010;376:1670–1681.
205. Besseling J, Hovingh GK, Huijgen R, et al. Statins in familial hypercholesterolemia: consequences for coronary artery disease and all-cause mortality. *J Am Coll Cardiol*. 2016;68:252–260.
206. Verschmissen J, Oosterveer DM, Yazdanpanah M, et al. Efficacy of statins in familial hypercholesterolaemia: a long-term cohort study. *BMJ*. 2008;337:a2423.
207. Perak AM, Ning H, de Ferranti SD, et al. Long-term risk of atherosclerotic cardiovascular disease in US adults with the familial hypercholesterolemia phenotype. *Circulation*. 2016;134:9–19.
208. DeFilippis AP, Young R, Carrubba CJ, et al. An analysis of calibration and discrimination among multiple cardiovascular risk scores in a modern multiethnic cohort. *Ann Intern Med*. 2015;162:266–275.
209. Samarasekera EJ, Neilson JM, Warren RB, et al. Incidence of cardiovascular disease in individuals with psoriasis: a systematic review and meta-analysis. *J Invest Dermatol*. 2013;133:2340–2346.
210. Feinstein MJ, Nance RM, Drozd DR, et al. Assessing and refining myocardial infarction risk estimation among patients with human immunodeficiency virus: a study by the Centers for AIDS Research Network of Integrated Clinical Systems. *JAMA Cardiol*. 2017;2:155–162.
211. Rubak S, Sandbaek A, Lauritzen T, et al. Motivational interviewing: a systematic review and meta-analysis. *Br J Gen Pract* 2005;55:305–312.
212. Artinian NT, Fletcher GF, Mozaffarian D, et al. Interventions to promote physical activity and dietary lifestyle changes for cardiovascular risk factor reduction in adults: a scientific statement from the American Heart Association. *Circulation* 2010;122:406–441.
213. Balady GJ, Williams MA, Ades PA, et al. American Heart Association Exercise CR. Core components of cardiac rehabilitation/secondary prevention programs: 2007 update: a scientific statement from the American Heart Association Exercise, Cardiac Rehabilitation, and Prevention Committee, the Council on Clinical Cardiology; the Councils on Cardiovascular Nursing, Epidemiology and Prevention, and Nutrition, Physical Activity, and Metabolism; and the American Association of Cardiovascular and Pulmonary Rehabilitation. *Circulation* 2007;115:2675–2682.
214. Piepoli MF, Corra U, Benzer W, et al. Secondary prevention through cardiac rehabilitation: from knowledge to implementation. A position paper from the Cardiac Rehabilitation Section of the European Association of Cardiovascular Prevention and Rehabilitation. *Eur J Cardiovasc Prev Rehabil* 2010;17:1–17.
215. Antithrombotic Trialists' (ATT) Collaboration, Baigent C, Blackwell L, et al. Aspirin in the primary and secondary prevention of vascular disease: collaborative meta-analysis of individual participant data from randomized trials. *Lancet*. 2009;373:1849–1860.
216. Ogawa H, Nakayama M, Morimoto T, et al. Low-dose aspirin for primary prevention of atherosclerotic events in patients with type 2 diabetes: a randomized controlled trial. *JAMA*. 2008;300:2134–2141.

217. ASCEND Study Collaborative Group, Bowman L, Mafham M, et al. Effects of aspirin for primary prevention in persons with diabetes mellitus. *N Engl J Med*. 2018;379:1529–1539.
218. Gaziano JM, Brotons C, Coppolecchia R, et al. Use of aspirin to reduce risk of initial vascular events in patients at moderate risk of cardiovascular disease (ARRIVE): a randomized, double-blind, placebo-controlled trial. *Lancet*. 2018;392:1036–1046.
219. Guirguis-Blake JM, Evans CV, Senger CA, et al. Aspirin for the primary prevention of cardiovascular events: a systematic evidence review for the U.S. Preventive Services Task Force. *Ann Intern Med*. 2016;164:804–813.
220. Raju N, Sobieraj-Teague M, Bosch J, et al. Updated meta-analysis of aspirin in primary prevention of cardiovascular disease. *Am J Med*. 2016;129:e35–e36.
221. Miedema MD, Duprez DA, Misialek JR, et al. Use of coronary artery calcium testing to guide aspirin utilization for primary prevention: estimates from the multi-ethnic study of atherosclerosis. *Circ Cardiovasc Qual Outcomes*. 2014;7:453–460.
222. Ridker PM. Should aspirin be used for primary prevention in the post-statin Era? *N Engl J Med*. 2018;379:1572–1574.
223. McNeil JJ, Wolfe R, Woods RL, et al. Effect of aspirin on cardiovascular events and bleeding in the healthy elderly. *N Engl J Med*. 2018;379:1509–1518.
224. García Rodríguez LA, Martín-Pérez M, Hennekens CH, et al. Bleeding risk with long-term low dose aspirin: a systematic review of observational studies. *PLoS ONE*. 2016;11:e0160046.
225. Darby SC, Ewertz M, McGale P, et al. Risk of ischemic heart disease in women after radiotherapy for breast cancer. *N Engl J Med* 2013;368:987–998.
226. Chen J, Long JB, Hurria A, et al. Incidence of heart failure or cardiomyopathy after adjuvant trastuzumab therapy for breast cancer. *J Am Coll Cardiol* 2012;60:2504–2512.
227. Darby S, McGale P, Peto R, et al. Mortality from cardiovascular disease more than 10 years after radiotherapy for breast cancer: nationwide cohort study of 90 000 Swedish women. *BMJ* 2003;326:256–257.
228. Plana JC, Galderisi M, Barac A, et al. Expert consensus for multimodality imaging evaluation of adult patients during and after cancer therapy: a report from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *Eur Heart J Cardiovasc Imaging* 2014;15:1063–1093.
229. McMurray JJ, Adamopoulos S, Anker SD, et al. ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure 2012: the Task Force for the Diagnosis and Treatment of Acute and Chronic Heart Failure 2012 of the European Society of Cardiology. Developed in collaboration with the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 2012;33:1787–1847.
230. Kalam K, Marwick TH. Role of cardioprotective therapy for prevention of cardiotoxicity with chemotherapy: a systematic review and meta-analysis. *Eur J Cancer* 2013;49:2900–2909.

Appendix

Agatston score 13
Albiglutide 27
Angiotensin converting enzyme inhibitors 28,29,36
Angiotensin receptor blockers 28,29,30
Ankle-brachial index 5,14,37
Ankylosing spondylitis 13
Anthracyclines 36
Anxiety 10,33
Aortic pulse wave velocity 19
Apolipoproteins 5,20,21
Arterial augmentation index 19
Arterial stiffness 5,19
Aspirin 28,35,36
Atorvastatin 27
Atrial fibrillation 5,6,12,15,18,20,26,29
Atrial flutter 15
Atrio-ventricular block 15
Bariatric surgery 26
Beta blockers 29,30,36
Body mass index (BMI) 26
Bupropion 25
Calcium channel blockers 29,30
Canagliflozin 27
Carotid intima-media thickness 5,14
Carotid ultrasound 6,14
Chemotherapy 6,35,36
Chronic kidney disease (CKD) 6,11,12,29,30,35,36
Computed tomography coronary angiography 6,19
Coronary artery disease (CAD) 5,6,10,11,12,16,17,18,19,20,26,27,29,31,36
Coronary calcium score 6,13,34,35
Dapagliflozin 27
DASH trial 22
Depression 10,14,34,35
Diabetes 6,8,10,12,14,15,16,17,21,23,26,27,28,30,31,32,33,35
Diuretics 29
Duke score 16
Dyslipidaemia 6,26,31
Electrocardiogram (ECG) 6,15,16
Electronic cigarettes (e-cigarettes) 25
Empagliflozin 27
Environmental tobacco smoking 25
Epigenetics 18
Erectile dysfunction 6,14
Exercise electrocardiogram 6,16,24
Ezetimibe 27
Genotype testing 6,18
Gestational diabetes 11

Global risk score (GRS) 6,8,9,14,15,32,33,34
 GLP-1R agonists 27,29
 Haemoglobin A1c (HbA1c) 6,10,15,16,26,28
 Heart failure 6,12,18,26,27,28,29
 High-sensitivity C-reactive protein (hsCRP) 6,12,13,37
 Hypercholesterolaemia 6,8,10,11,14,21,30,31
 Hypertension 6,8,11,12,14,15,16,17,18,21,22,23,28,29,30,35,37
 Lipoprotein(a) 6,21
 Lipoprotein-associated phospholipase A2 6,19
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 Metformin 27,29
 Microalbuminuria 6,11,12,32,33
 Mono unsaturated fatty acids 22
 Myocardial infarction 6,8,13,14,16,18,26,27,31,35
 Myocardial perfusion imaging 6,17
 Natriuretic peptides 6,18,37
 Nicotine replacement therapy 25
 Non-steroidal anti-inflammatory drugs 35
 Novacode criteria 15
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 Obstructive sleep apnoea 6,18
 Orlistat 26
 Passive smoking 25
 Peripheral vascular disease 12,14,29
 Polycystic ovarian syndrome 11,37
 Polyunsaturated fatty acids 22
 Positron emission tomography (PET) 17
 Preeclampsia 11,37
 Pregnancy induced hypertension 11,37
 Premature menopause 11
 Psoriasis 13,33
 Radiotherapy 36
 Rheumatoid arthritis 13,14,33
 Schizophrenia 10
 Semaglutide 27
 SGLT2 inhibitors 27,29
 Simvastatin 27
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 Statins 28,29,31,36
 Steroids 35
 Stress 10,34
 Stress echocardiography 6,20
 Stroke 8,11,12,14,18,21,22,26,27,28,29,35
 Sudden cardiac death 12
 Supra ventricular tachycardia 15
 Systemic lupus erythematosus 13

Trans fatty acids 22
Transthoracic echocardiography 6,16,17
Trastuzumab 36
Varenicline 25
Ventricular tachycardia 15,20
Waist circumference 26
Waist to hip ratio 26
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Supplementary Data and Links

1. 2019 ACC/AHA Guideline on the Primary Prevention of Cardiovascular Disease: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. J Am Coll Cardiol. 2019 Mar, 74 (10) e177–e232. https://www.jacc.org/doi/full/10.1016/j.jacc.2019.03.010?_ga=2.12142897.408972830.1604755770-1634440621.1486620085
2. 2016 European Guidelines on cardiovascular disease prevention in clinical practice: The Sixth Joint Task Force of the European Society of Cardiology and Other Societies on Cardiovascular Disease Prevention in Clinical Practice. Developed with the special contribution of the European Association for Cardiovascular Prevention & Rehabilitation (EACPR), European Heart Journal, Volume 37, Issue 29, 1 August 2016, Pages 2315-2381. <https://doi.org/10.1093/eurheartj/ehw106>
3. 2010 ACCF/AHA Guideline for Assessment of Cardiovascular Risk in Asymptomatic Adults A Report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. Circulation. 2010;122:e584–e636.